

# IDENTIFICATION IN INSTRUMENTAL VARIABLES MODELS: THE CENTRAL ROLE OF ABADIE'S KAPPA

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We study instrumental variables models characterized by: (i) Unobserved heterogeneity consisting of potential outcomes and response types that describe how the instrument determines treatment choice; (ii) Conditional independence of the instrument and the unobserved heterogeneity; and (iii) Convex restrictions on the distribution of unobserved heterogeneity. We show certain causal parameters are identified in these models if and only if a version of the kappa of Abadie (2003) exists. Our identification results are constructive in yielding estimating moment conditions. Focusing on a leading special case, we develop asymptotically normal estimators based on a doubly robust version of these moment conditions.

KEYWORDS: Potential outcomes, instrumental variables, identification.

## 1. INTRODUCTION

Potential outcomes models are the leading framework for identifying and estimating causal effects with heterogeneous treatment responses. Originally developed for random-

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ized experiments by [Neyman \(1990\)](#), they were shown by [Rubin \(1974\)](#) to provide a foundation for causal inference in observational studies. Their central role in instrumental variables models was established by [Imbens and Angrist \(1994\)](#) who showed the importance of restricting the manner in which an instrument determines treatment choice. A subsequent literature has built on this work by developing identifying restrictions for a wide range of empirically relevant settings, including applications with ordered and unordered treatments, multiple instruments, and mediating variables.

In this paper, we propose a class of instrumental variables models that unifies and expands upon these identification strategies. The core assumptions of our framework are: (i) Unobserved heterogeneity consists of potential outcomes and response types that describe how the instrument determines treatment choice; (ii) Instrumental variables are conditionally independent of unobserved heterogeneity in both potential outcomes and response types; and (iii) The distribution of unobserved heterogeneity belongs to a convex set. Models within the scope of this framework include those that secure identification by imposing assumptions that restrict the possible response types or, in the terminology of [Frangakis and Rubin \(2002\)](#), principal strata. These encompass, among many others, the monotonicity restriction of [Angrist and Imbens \(1995\)](#), the partial monotonicity requirement of [Mogstad et al. \(2021\)](#), and the revealed preference restrictions of [Kline and Walters \(2016\)](#) and [Pinto \(2021\)](#). Additional examples of identifying assumptions that fall within our framework include the sequential ignorability requirement of [Imai et al. \(2010\)](#) and the comparative compliers assumption of [Mountjoy \(2022\)](#).

Our main contribution is to show that these models share a common necessary and sufficient condition for identifying causal parameters that can be expressed as expectations of functions of unobserved heterogeneity and covariates. Intuitively, our results show that identification holds if and only if a suitable generalization of the kappa of [Abadie \(2003\)](#) exists. More formally, we show that identification is equivalent to the function whose expectation we seek belonging to the closure of the range of a linear operator  $\Upsilon$  that maps functions of observables to functions of unobservables. Identification is therefore tantamount to the existence of a sequence of functions  $\{\kappa_j\}$  of observable variables such that  $\Upsilon(\kappa_j)$  suitably approximates the function of unobserved heterogeneity whose expectation we wish to identify. Critically, the map  $\Upsilon$  is the same across all the models in our framework, but the sense in which  $\Upsilon(\kappa_j)$  must provide a suitable approximation depends on the

restrictions imposed by the specific model. The functions  $\{\kappa_j\}$  may be viewed as a generalization of Abadie’s kappa, and our results can therefore be interpreted as showing that the identification strategy of Abadie (2003) is central to identifying causal parameters in a class of instrumental variable models that extends beyond Imbens and Angrist (1994).

Our identification result is constructive. We show that if the parameter of interest is identified, then it equals the limit of the expectations of the approximating sequence  $\{\kappa_j\}$ . This characterization suggests a simple estimation strategy: Estimate the relevant  $\{\kappa_j\}$  and compute their sample averages. Establishing asymptotic normality when the nuisance parameters  $\{\kappa_j\}$  are estimated using machine learning methods, however, often requires characterizing orthogonal scores that depend on the specific restrictions being imposed by the model. For estimation, we therefore focus on a leading special case of our framework in which the identifying restrictions do not depend on the distribution of observables. In this context, we derive a double robust moment condition and follow Smucler et al. (2019) and Chernozhukov et al. (2022a,b) by employing  $\ell_1$ -regularization to estimate nuisance parameters. We show that the resulting estimators are asymptotically normally distributed and that bootstrap based inference is asymptotically valid. We provide code for assessing whether a parameter is identified and implementing our estimator when it is for the special case of applications with discrete instruments and restriction that only limit response types.<sup>1</sup>

When specialized to existing models within our framework, our results both recover existing results and generate new insights. For example, in the setting of Imbens and Angrist (1994), our identification results recover Abadie (2003), while our estimation results yield the double robust moment condition used by Tan (2006) and Singh and Sun (2022). In the related discrete instrument models studied by Heckman and Pinto (2018), our results provide conditions under which an assumption they show to be sufficient for identification is also necessary.<sup>2</sup> In other applications, our results can characterize what parameters are identified by existing assumptions. Mogstad et al. (2018), for example, introduce a partial monotonicity assumption in order to interpret the two stage least squares estimand causally.

<sup>1</sup>Found at the following public GitHub repository: <https://github.com/mnavjeev/IdentificationCheck.git>.

<sup>2</sup>Corollary C-1 in Heckman and Pinto (2018) provides necessary and sufficient conditions for linear restrictions implied by the model to deliver identification. Our results in contrast provide necessary and sufficient conditions for identification that reflect all the restrictions of the model.

As an illustration, we show how our results extend their analysis by readily characterizing all other causal estimands that are identified by the same partial monotonicity requirement.

Our results are not only useful in the context of existing models. By providing a transparent connection between assumptions and the parameters they identify, our results are also helpful for developing, identifying, and estimating new causal models. We illustrate this point with an empirical re-analysis of the Moving to Opportunity (MTO) experiment (Orr et al., 2003, Kling et al., 2007, Orr, 2011). Specifically, we evaluate a conjecture by Ludwig et al. (2008) who suggested that improved mental health may play an important role in the causal mechanism through which moving from high to low-poverty neighborhoods impacts economic outcomes. Guided by our necessary and sufficient conditions for identification, we devise a model that enables us to identify and estimate the mediating effects of improved mental health for different subpopulations. The resulting model is new and distinct from the mediation analysis in Frölich and Huber (2017), Ding and Lu (2017), and Rudolph et al. (2024).

This paper contributes to the literature on potential outcomes models. Our analysis appears to be the first to establish the unifying role of a common necessary and sufficient condition for identification. In independent and subsequent work, Goff (2025) studies necessary and sufficient conditions for identifying treatment effects conditional on response type in the discrete instrument model of Heckman and Pinto (2018). Angrist et al. (2026) builds on our results by introducing new causal parameters and deriving transparent necessary and sufficient conditions for their identification in applications with a single Bernoulli instrument. Our analysis is further related to work providing analytical (Manski, 1990) or computational (Balke and Pearl, 1997, Mogstad et al., 2018) bounds for partially identified parameters. By showing that point identification turns on a single linear equation, our results provide a simple way to determine when such bounds collapse to a point.

The rest of the paper is organized as follows. Section 2 introduces the class of models we study, and Section 3 illustrates our results in the context of the MTO experiment. Our main theoretical results are in Sections 4 and 5, which discuss identification and estimation.

## 2. THE MODEL

We consider applications with a scalar outcome  $Y \in \mathbf{Y} \subseteq \mathbf{R}$ , a discrete treatment  $T \in \mathbf{T} \equiv \{t_1, \dots, t_d\}$ , an instrument  $Z \in \mathbf{Z}$ , and covariates  $X \in \mathbf{X}$ . The instrument  $Z$  and

covariates  $X$  may be vectors (e.g.,  $\mathbf{Z}, \mathbf{X} \subseteq \mathbf{R}^k$ ) or, as we discuss in the appendix, have a more general structure. We model unobserved heterogeneity through the potential outcomes  $Y^* \equiv (Y^*(t_1), \dots, Y^*(t_d))$  and a response type  $T^* : \mathbf{Z} \rightarrow \{t_1, \dots, t_d\}$  that describes how the instrument determines a unit's treatment decision. The observed treatment and outcome are given by  $T = T^*(Z)$  and  $Y = Y^*(T)$ .

In order to state the assumptions that characterize our model, we first define

$$L^p(Q) \equiv \{f : \|f\|_{Q,p} < \infty\} \quad \|f\|_{Q,p} \equiv \left( \int |f|^p dQ \right)^{\frac{1}{p}};$$

i.e.,  $L^p(Q)$  denotes the set of functions that have a finite  $p^{th}$  moment under  $Q$ . We say that a distribution  $Q$  is absolutely continuous with respect to (w.r.t.) a distribution  $Q'$ , denoted  $Q \ll Q'$ , if  $Q$  assigns zero probability to any event to which  $Q'$  assigns zero probability. Whenever  $Q \ll Q'$ ,  $Q$  has a density w.r.t.  $Q'$  that we denote by  $dQ/dQ'$ . We also let  $Q_0$  denote the distribution of  $(Y^*, T^*, Z, X)$  and  $P$  the distribution of  $(Y, T, Z, X)$ .

Given the introduced notation, we impose the following two assumptions.

**ASSUMPTION 2.1:** (i)  $(Y^*, T^*, Z, X) \sim Q_0$  with  $Y^* \equiv (Y^*(t_1), \dots, Y^*(t_d)) \in \mathbf{Y}^* \subseteq \mathbf{R}^d$ ,  $Z \in \mathbf{Z}$ ,  $X \in \mathbf{X}$ , and  $T^* \in \mathbf{T}^*$  with  $\mathbf{T}^*$  a set of functions from  $\mathbf{Z}$  to  $\mathbf{T} \equiv \{t_1, \dots, t_d\}$ ; (ii) We observe  $T = T^*(Z)$ ,  $Y = Y^*(T)$ ,  $Z$ , and  $X$  with  $(Y, T, Z, X) \sim P$ .

**ASSUMPTION 2.2:** (i)  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $Q_0$ ; (ii)  $Q_0 \ll \mu$  for some separable probability measure  $\mu$ ; (iii)  $dQ_0/d\mu$  belongs to a set  $\mathcal{Q} \subseteq L^1(\mu)$  for  $\mathcal{Q}$  a closed convex subset of Banach Space  $(\mathbf{Q}, \|\cdot\|_{\mathbf{Q}})$  with  $\|\cdot\|_{\mathbf{Q}}$  (weakly) stronger than  $\|\cdot\|_{\mu,1}$ .

Assumption 2.1 formalizes our setup. Assumption 2.2(i) requires that  $Z$  be exogenous, while Assumption 2.2(ii) imposes that the distribution of  $(Y^*, T^*, Z, X)$  be absolutely continuous with respect to a measure  $\mu$ . A choice of  $\mu$  corresponds to a choice of support restrictions imposed by the model. For example, since  $Q_0$  must assign zero probability to any event to which  $\mu$  assigns zero probability, we may employ  $\mu$  to rule out certain realizations of  $T^*$ , such as defiers in Imbens and Angrist (1994). Assumption 2.2(iii) accommodates additional convex restrictions. These restrictions may include regularity conditions that ensure the parameter of interest is well-defined (see Example 2.1) and more substantive identifying assumptions (see Section 3). We note that  $\mu$ ,  $\mathcal{Q}$ , and  $\mathbf{Q}$  may all depend on  $P$ , though do not make the dependence explicit – i.e., they may be identified but not known.

We also highlight that different measures  $\mu$  can yield the same model by, e.g., imposing the same support restrictions through Assumption 2.2(ii). This flexibility is often helpful, as it allows us to select measures  $\mu$  that simplify calculations; see Remark 4.1.

The distribution  $Q_0$  of  $(Y^*, T^*, Z, X)$  induces, through Assumption 2.1, the distribution  $P$  of  $(Y, T, Z, X)$ . Absent restrictive assumptions,  $Q_0$  is not identified because there are alternative distributions  $Q$  for  $(Y^*, T^*, Z, X)$  that induce  $P$ . We refer to any such distribution  $Q$  as observationally equivalent to  $Q_0$ . The identified set for  $Q_0$  is defined as

$$\Theta_0 \equiv \{Q : Q \text{ is obs. equiv. to } Q_0, (Y^*, T^*) \perp\!\!\!\perp Z|X \text{ under } Q, Q \ll \mu, \frac{dQ}{d\mu} \in \mathcal{Q}\};$$

i.e.,  $\Theta_0$  is the set of distributions for  $(Y^*, T^*, Z, X)$  that induce  $P$  and satisfy the requirements imposed on  $Q_0$  in Assumption 2.2.

Our primary goal is to study the identification and estimation of features of the true distribution  $Q_0$ . Concretely, for any given function  $\ell$  of  $(Y^*, T^*, X)$  we let

$$\lambda_Q \equiv E_Q[\ell(Y^*, T^*, X)], \quad (1)$$

where the  $Q$  subscript emphasizes that the expectation is taken with respect to a distribution  $Q$  that may not equal  $Q_0$ . We study identification of the expectation of  $\ell(Y^*, T^*, X)$  under  $Q_0$ , which we denote by  $\lambda_{Q_0}$ . This expectation is identified if and only if  $\lambda_Q$  takes the same value for every  $Q \in \Theta_0$ . We emphasize that  $\ell$  can depend on  $P$ , though we leave such dependence implicit – i.e.,  $\ell$  must be an identified function but not necessarily known.

## 2.1. Examples

In order to fix ideas, we introduce examples that highlight the flexibility of our setup.

**EXAMPLE 2.1:** Following Rosenbaum and Rubin (1983), suppose treatment  $T \in \{0, 1\}$  is Bernoulli and  $Y^* \equiv (Y^*(0), Y^*(1))$  is independent of  $T$  conditional on  $X$ . To map this setting into our framework, let  $Z = T$ , select  $\mu$  to satisfy  $\mu(T^*(Z) = Z) = 1$ , and note Assumption 2.2(i) is then equivalent to  $Y^* \perp\!\!\!\perp T|X$ . The identified set  $\Theta_0$  consists of the distributions  $Q$  for  $(Y^*, T, X)$  that induce the distribution  $P$  of observables and satisfy the maintained assumptions, while  $Q_0$  denotes the true distribution of  $(Y^*, T, X)$ . Setting  $\ell(Y^*, T^*, X) = Y^*(1) - Y^*(0)$ , we can write the average treatment effect (ATE) as the

expectation of  $\ell$  under  $Q_0$ . In this model, the ATE is identified, which is equivalent to  $\lambda_Q \equiv E_Q[Y^*(1) - Y^*(0)]$  being constant in  $Q \in \Theta_0$ . Ensuring the ATE is well-defined requires us to assume  $(Y^*(0), Y^*(1))$  has a first moment, which can be imposed through Assumption 2.2(iii). Other choices of  $\ell$ , such as  $\ell(Y^*, T^*, X) = 1\{Y^*(1) \geq Y^*(0)\}$ , lead to parameters  $\lambda_{Q_0}$  that are not identified because  $\lambda_Q$  changes with  $Q \in \Theta_0$ . ■

EXAMPLE 2.2: Consider a special case of Imbens and Angrist (1994) in which  $T \in \{0, 1\}$  and  $Z \in \{0, 1\}$  are Bernoulli. In this context,  $T^*$  maps  $\mathbf{Z} \equiv \{0, 1\}$  to  $\mathbf{T} \equiv \{0, 1\}$ . Assumption 2.2(ii) allows us to impose that the instrument weakly induce individuals into treatment by setting  $\mu$  to satisfy  $\mu(T^*(1) \geq T^*(0)) = 1$ . If we let  $\ell(Y^*, T^*, X) = 1\{T^*(1) > T^*(0)\}$ , then  $\lambda_Q$  equals the probability of compliers under any distribution  $Q$  for  $(Y^*, T^*, Z, X)$ , while  $\lambda_{Q_0}$  equals the true probability of compliers. The probability of compliers is identified because  $\lambda_Q$  is constant in  $Q \in \Theta_0$ . To study parameters such as the marginal distribution of  $Y^*$  for compliers, as in Imbens and Rubin (1997), we write

$$E_{Q_0}[f(Y^*(t))|T^*(1) > T^*(0)] = \frac{E_{Q_0}[f(Y^*(t))1\{T^*(1) > T^*(0)\}]}{E_{Q_0}[1\{T^*(1) > T^*(0)\}]} \quad (2)$$

for any given function  $f$  and  $t \in \{0, 1\}$ . The conditional expectation in (2) equals the ratio of two parameters with the structure in (1) with  $\ell$  known. Our results provide necessary and sufficient conditions for the identification of the numerator, the denominator, and also the ratio in (2); see Remark 4.3 for details. ■

EXAMPLE 2.3: Heckman and Vytlacil (2005) study a model in which treatment is determined by  $T = 1\{f(X, Z) \geq \xi\}$  with  $f$  an unknown continuous function,  $\xi$  unobservable, and  $(Y^*(0), Y^*(1), \xi) \perp\!\!\!\perp Z|X$ . Assuming that  $Z$  is a scalar and  $f(X, \cdot)$  is increasing, we may map this model into our framework by letting  $T^* \equiv 1\{f(X, \cdot) \geq \xi\}$  and imposing<sup>3</sup>

$$\mu(\lim_{z \downarrow z'} T^*(z) = T^*(z') \text{ and } T^*(z) \geq T^*(z') \text{ for all } z \geq z') = 1.$$

The literature has focused on causal parameters that can be expressed as expectations of  $h(Y^*, \xi, X)$  for some function  $h$ . Because  $T^*$  may not be an invertible function of  $\xi$ , these

<sup>3</sup>The upper semicontinuity of  $T^*$  is due to the continuity of  $f(X, \cdot)$ . The case in which  $f(X, \cdot)$  is not monotonic corresponds to imposing  $T^*(z) \geq T^*(z')$  whenever  $P(T = 1|Z = z, X) \geq P(T = 1|Z = z', X)$ .

expectations may not map into the parameters in (1) that we study. However, under regularity conditions, it is possible to show that a necessary condition for the expectation of  $h$  to be identified is that  $h$  depend on  $\xi$  only through  $T^*$ . Hence, our characterization of identification of (1) characterizes identification of causal parameters that can be expressed as expectations of  $h(Y^*, \xi, X)$  for some function  $h$  as well. Similarly, our framework can accommodate related structural models, such as those in [Lee and Salanié \(2018\)](#). ■

EXAMPLE 2.4: [Kline and Walters \(2016\)](#) evaluate the cost-effectiveness of the Head Start Program. In their analysis,  $Z \in \{0, 1\}$  denotes whether an individual was offered to attend a Head Start school and  $T$  can take three values: Attend a Head Start School ( $h$ ), attend other schools ( $c$ ), or receive home care ( $n$ ). Here,  $T^*$  maps  $\{0, 1\}$  to  $\{h, c, n\}$  and we can therefore characterize  $T^*$  as a vector  $T^* = (T^*(0), T^*(1))$  taking values in  $\{h, c, n\} \times \{h, c, n\}$ . The main identification assumption in [Kline and Walters \(2016\)](#) is that receiving an offer to attend a Head Start school may only induce individuals to attend a Head Start school. Formally, they require that  $T^* = (T^*(0), T^*(1))$  belong to

$$\mathbf{R}^* \equiv \left\{ \begin{pmatrix} n \\ h \end{pmatrix}, \begin{pmatrix} c \\ h \end{pmatrix}, \begin{pmatrix} n \\ n \end{pmatrix}, \begin{pmatrix} c \\ n \end{pmatrix}, \begin{pmatrix} h \\ h \end{pmatrix} \right\},$$

which can be mapped into our framework by setting  $\mu$  to satisfy  $\mu(T^* \in \mathbf{R}^*) = 1$ . More generally, in applications with a discrete instrument we may always impose support restrictions on  $T^*$  by demanding that  $\mu(T^* \in \mathbf{R}^*) = 1$  for some finite set of vectors  $\mathbf{R}^* \equiv \{t_1^*, \dots, t_r^*\}$ . Through this observation, our framework accommodates the unordered monotonicity condition of [Heckman and Pinto \(2018\)](#), the analysis of the MTO experiment by [Pinto \(2021\)](#), and the survey non-response model of [Dutz et al. \(2021\)](#). ■

EXAMPLE 2.5: [Mogstad et al. \(2021\)](#) propose a partial monotonicity condition that yields a causal interpretation for two stage least squares (TSLS) in applications with multiple instruments. In an empirical re-examination of [Carneiro et al. \(2011\)](#), the authors consider a setting in which  $T \in \{0, 1\}$  indicates whether an individual attended college,  $Y$  represents log average hourly wage, and  $Z = (C, W)$  with  $C \in \{0, 1\}$  indicating whether a college is present in the county of residence at age 14 and  $W$  denoting average log earnings in the county of residence at age 17. [Mogstad et al. \(2021\)](#) suppose that increasing  $C$  induces individuals into treatment, while increasing  $W$  induces individuals out of treatment.

Their requirement may be mapped into our framework by selecting  $\mu$  to satisfy

$$\mu(T^*(1, w) \geq T^*(0, w) \text{ and } T^*(c, w) \leq T^*(c, w') \text{ for all } c \in \{0, 1\}, w \geq w') = 1. \quad (3)$$

Restrictions on the response types that are analogous to (3) were also employed in the empirical study of the returns to two-year colleges by [Mountjoy \(2022\)](#). ■

EXAMPLE 2.6: Mediation analysis aims to identify how a treatment affects an outcome through intermediate variables called mediators. [Angrist et al. \(2022\)](#), for instance, hypothesize that engagement in the first year of college mediates the impact of student grants on graduation rates. Letting  $D \in \{0, 1\}$  indicate whether a student is awarded a grant,  $M \in \{e, ne\}$  whether she was engaged ( $e$ ) or not ( $ne$ ), and  $Y \in \{0, 1\}$  whether she graduated college, we may map their study into our framework by letting  $T = (D, M)$  and  $Z = D$ . Potential outcomes  $Y^*$  are then indexed by  $t = (d, m) \in \{0, 1\} \times \{e, ne\}$ , while  $T^*$  is a function mapping  $\mathbf{Z} \equiv \{0, 1\}$  to  $\mathbf{T} \equiv \{0, 1\} \times \{e, ne\}$ . Assumptions 2.2(i)(ii) can then be employed to impose, for example, the sequential ignorability requirement of [Imai et al. \(2010\)](#). [Angrist et al. \(2026\)](#) build on the mediation model we develop in Section 3 to assess the mediating role of early engagement discussed in [Angrist et al. \(2022\)](#). ■

### 3. MOVING TO OPPORTUNITY

As a preview of our results, we first illustrate the ability of our methods to develop, identify, and estimate a causal model in the context of an empirical analysis of MTO. MTO was an experiment in which households living in high-poverty neighborhoods were offered vouchers that incentivized them to relocate to low-poverty neighborhoods. The intervention generated substantial and long-lasting improvements in mental health but more nuanced effects on economic outcomes ([Katz et al., 2001](#), [Ludwig et al., 2012](#), [Pinto, 2021](#)).

While mental health and economic outcomes may be mutually causally related, [Kling et al. \(2004\)](#) find reduced stress due to relocation is the primary cause of improved mental health, rather than improvements in economic outcomes. Building on these results, [Ludwig et al. \(2008\)](#) conjecture that improved mental health may be an important mediator through which the MTO intervention affects labor market participation. In what follows, we revisit the MTO experiment to examine this hypothesis.

We use data on mental health and labor market outcomes collected in the Interim Evaluation (2001-2002, 4-7 years after randomization) and control for baseline variables surveyed at the start of the intervention. We let  $Z \in \{0, 1\}$  indicate whether a voucher was offered,  $Y$  denote an outcome of interest, and  $D \in \{0, 1\}$  indicate whether the household relocated to a low-poverty neighborhood. The measure of mental health is an indicator,  $M \in \{0, 1\}$ , of whether the sampled adult reported feeling calm and peaceful all or most of the time during the previous thirty days. We set  $T = (D, M)$  and let  $Y^*(t)$  depend on  $t = (d, m)$  to reflect that mental health and relocation can affect outcomes. Covariates,  $X$ , consist of experimental site indicators as well as household and neighborhood characteristics.

### 3.1. Learning About Types

We begin by studying features of the distribution of response types  $T^*$ . In particular, for functions  $\ell$  of  $(T^*, X)$ , we first identify and estimate parameters with the structure

$$E_{Q_0}[\ell(T^*, X)]. \quad (4)$$

By selecting  $\mu$  in Assumption 2.2(ii) to restrict the support of  $T^*$ , our model enables us to restrict how a voucher offer affects relocation decisions and mental health. Under such support restrictions, our results imply that (4) is identified if and only if there exists a function  $\nu$  of  $(T, Z, X)$  such that for all  $t^*$  in the support of  $T^*$  and all  $X$  we have

$$\sum_{t \in \mathbf{T}} \sum_{z \in \mathbf{Z}} 1\{t^*(z) = t\} \nu(t, z, X) = \ell(t^*, X); \quad (5)$$

see Corollary 4.3. Any solution  $\nu$  to (5) can be used to identify (4) by constructing the function  $\kappa(t, z, X) = \nu(t, z, X)/P(Z = z|X)$ , which our results show satisfies

$$E_{Q_0}[\ell(T^*, X)] = E_P[\kappa(T, Z, X)]. \quad (6)$$

Guided by this result, we impose three requirements that deliver identification of the distribution of  $(T^*, X)$ : (i) A voucher offer (weakly) incentivizes households to relocate; (ii) Moving to a low-poverty neighborhood (weakly) improves mental health; and (iii) A voucher offer affects mental health only through the relocation decision. Condition (ii) is motivated by extensive empirical work finding statistically significant estimates of the causal effect of relocation on mental health in the MTO experiment (Kling et al., 2004).

Condition (ii) may nonetheless be violated, for instance, if there are individuals for whom the move causes social displacement that outweighs the benefits on mental health from moving to a lower poverty and safer neighborhood. Condition (iii) can be problematic for individuals who would have moved without a voucher, and for whom the voucher offer therefore represents a positive income shock. This concern is partly allayed by the estimated proportion of such individuals in our sample being below 2.5% (see Table I below).

We impose these restrictions by letting  $T^*(z) \equiv (D^*(z), M^*(z))$  with  $D^*$  and  $M^*$  describing how relocation and mental health respond to a voucher offer, and setting

$$\mu(D^*(1) \geq D^*(0), M^*(1) \geq M^*(0), \text{ and } D^*(1) = D^*(0) \Rightarrow M^*(1) = M^*(0)) = 1. \quad (7)$$

These restrictions limit the support of  $T^*$  to seven possible types. These types, displayed in the first panel of Table I, are characterized by the possible realizations of  $D^*$  and  $M^*$  – i.e. by whether they are never takers, compliers, or always takers with regard to relocation and mental health status. For instance, types CN, CA, and CC are compliers with respect to relocation in that they relocate if and only if they are offered a voucher.

For any given response type  $t_0^*$ , we can write the probability that  $T^*$  equals  $t_0^*$  as the expectation in (4) evaluated at  $\ell(T^*, X) = 1\{T^* = t_0^*\}$ . Setting  $\ell(t^*, X) = 1\{t^* = t_0^*\}$  in (5) yields a system of linear equations (in  $\nu$ ) that can be verified to have a solution for any  $t_0^*$  in the support of  $T^*$ .<sup>4</sup> Hence, the probability of each type is identified, and we can apply our asymptotically normal estimator based on (6) to estimate it; see Theorem 5.1. Table I reports these estimates. While all types in our model occur with a strictly positive probability, the vast majority of households either do not relocate with a voucher offer (types NN, NA) or only relocate when offered a voucher (types CN, CC, CA).

Type CC is the only response type that experiences mental health improvement after relocation. Our results allow us to estimate how CC differs along baseline characteristics.<sup>5</sup> These estimates, reported in Table I, suggest that type CC is less attached to the baseline

<sup>4</sup>Let  $\mathbf{T} = \{t_1, \dots, t_4\} = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$  and set  $\omega_0(t^*) = (1\{t^*(0) = t_1\}, \dots, 1\{t^*(0) = t_4\})'$  and  $\omega_1(t^*) = (1\{t^*(1) = t_1\}, \dots, 1\{t^*(1) = t_4\})'$ . For any  $t_0^*$  we can find vectors  $s_0$  and  $s_1$  solving  $s_0' \omega_0(t^*) + s_1' \omega_1(t^*) = 1\{t^* = t_0^*\}$  for all  $t^*$  by, e.g., linear regression. Letting  $(\nu(t_1, z), \dots, \nu(t_4, z)) = s_z'$  then solves (5).

<sup>5</sup>E.g., since the probability  $P(T^* = \text{CC})$  is identified, we may set  $\ell(T^*, X) = X 1\{T^* = \text{CC}\} / P(T^* = \text{CC})$  in which case (4) becomes the mean of  $X$  conditional on being type CC.

TABLE I  
TYPE PROBABILITIES AND CONDITIONAL EXPECTATION OF BASELINE VARIABLES

|                                             | Type Definitions and Probabilities               |                  |                  |                  |                  |                  |                  |
|---------------------------------------------|--------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                             | NN                                               | NA               | CN               | CC               | CA               | AN               | AA               |
| $(D^*(0), M^*(0))$                          | (0,0)                                            | (0,1)            | (0,0)            | (0,0)            | (0,1)            | (1,0)            | (1,1)            |
| $(D^*(1), M^*(1))$                          | (0,0)                                            | (0,1)            | (1,0)            | (1,1)            | (1,1)            | (1,0)            | (1,1)            |
| Probability Point Estimate                  | 0.259                                            | 0.253            | 0.196            | 0.068            | 0.201            | 0.012            | 0.010            |
| s.e.                                        | (0.014)                                          | (0.014)          | (0.013)          | (0.024)          | (0.023)          | (0.004)          | (0.003)          |
|                                             | Expected Value of Covariate Conditional on Types |                  |                  |                  |                  |                  |                  |
|                                             | NN                                               | NA               | CN               | CC               | CA               | AN               | AA               |
| Household member<br>has a disability        | 0.184<br>(0.021)                                 | 0.145<br>(0.019) | 0.154<br>(0.024) | 0.382<br>(0.156) | 0.077<br>(0.044) | 0.097<br>(0.080) | 0.048<br>(0.049) |
| No friends in the<br>neighborhood           | 0.347<br>(0.028)                                 | 0.392<br>(0.030) | 0.422<br>(0.035) | 0.568<br>(0.187) | 0.409<br>(0.058) | 0.361<br>(0.132) | 0.775<br>(0.138) |
| Reports watching for<br>neighbor's children | 0.551<br>(0.029)                                 | 0.596<br>(0.031) | 0.509<br>(0.035) | 0.355<br>(0.191) | 0.559<br>(0.059) | 0.840<br>(0.090) | 0.514<br>(0.152) |

First panel reports the support points of  $T^*$  and their respective estimated probabilities. Second panel reports estimates for the expectation of baseline variables conditioned on types. All estimates account for the person-level weight for the adult survey of the interim analyses. Standard errors are displayed in parentheses.

neighborhood and therefore less likely to experience social displacement after moving. They are second only to type AA in reporting not having friends in their neighborhood (at 56.8%) and the least likely to report watching for neighbor's children (at 35.5%). They are also the most likely to have a household member with a disability (at 38.2%), which we may expect to be associated with incurring greater benefits from moving to a low poverty neighborhood. While the point estimates are suggestive, we note that we cannot statistically reject that these baseline characteristics for type CC are different from those of other types.

### 3.2. Learning About Outcomes

We next turn to estimating treatment effects in our model. A natural starting point is to examine the LATE of [Imbens and Angrist \(1994\)](#), which in our context equals

$$\text{LATE} \equiv E_{Q_0}[Y^*(1, M^*(1)) - Y^*(0, M^*(0)) | T^* \in \{\text{CN}, \text{CC}, \text{CA}\}].$$

The LATE informs us about the treatment effect of relocating for individuals who decide to relocate in response to a voucher offer. However, the LATE is a weighted average of causal effects across types with different mental health statuses and is, as a result, not suitable for assessing the mediating role of mental health. Specifically, the LATE aggregates

$$\begin{aligned} \text{CDE}_0 &\equiv E_{Q_0}[Y^*(1, 0) - Y^*(0, 0)|T^* = \text{CN}] \\ \text{CDE}_1 &\equiv E_{Q_0}[Y^*(1, 1) - Y^*(0, 1)|T^* = \text{CA}] \\ \text{CTE} &\equiv E_{Q_0}[Y^*(1, 1) - Y^*(0, 0)|T^* = \text{CC}]; \end{aligned} \quad (8)$$

i.e., the LATE is a weighted average of the “controlled direct effects” of relocating while keeping mental health status constant ( $\text{CDE}_0$  and  $\text{CDE}_1$ ) and the “controlled total effect” of simultaneously relocating and improving mental health (CTE).

Because the marginal distribution of  $T^*$  is identified, the identification of  $\text{CDE}_0$ ,  $\text{CDE}_1$ , and CTE reduces to the identification of expectations with the structure<sup>6</sup>

$$E_{Q_0}[\rho(Y^*(t))\ell(T^*, X)]. \quad (9)$$

Our results imply that restriction (7) does not identify  $\text{CDE}_0$ ,  $\text{CDE}_1$ , and CTE; see Corollary 4.4. We therefore introduce an “exogeneity of irrelevant mediator choices” (EIMC) assumption: Potential outcomes for high (resp. low) poverty neighborhood and poor (resp. good) mental health are conditionally independent of what mental health would have been in a low (resp. high) poverty neighborhood. Formally, we use Assumption 2.2(iii) to require

$$\begin{aligned} Y^*(0, 0) &\perp\!\!\!\perp M^*(1)|D^*(1) > D^*(0), M^*(0) = 0, X \\ Y^*(1, 1) &\perp\!\!\!\perp M^*(0)|D^*(1) > D^*(0), M^*(1) = 1, X. \end{aligned}$$

EIMC introduces some homogeneity among compliers by implying that, conditional on  $X$ , the distribution of  $Y^*(0, 0)$  is the same for types CC and CN, and the distribution of  $Y^*(1, 1)$  is the same for types CC and CA.<sup>7</sup> This assumption is a slight weakening of a

<sup>6</sup>E.g.,  $E[Y^*(t)|T^* = \text{CN}]$  equals (9) with  $\rho(Y^*(t)) = Y^*(t)$  and  $\ell(T^*, X) = 1\{T^* = \text{CN}\}/P(T^* = \text{CN})$ .

<sup>7</sup>E.g., in a threshold crossing model in which  $D^*(z) = 1\{z \geq \eta\}$  and  $M^*(z) = 1\{D^*(z) \geq \varepsilon\}$ , EIMC is implied by  $Y^*(t)$  being independent of  $\varepsilon$  conditionally on  $\eta$  and  $\varepsilon$  and  $\eta$  being independent.

TABLE II  
TREATMENT EFFECTS ESTIMATES

|                  | Sampled Adult<br>is Employed | Sampled Adult not<br>in Labor Force | Sampled Adult<br>Earnings | Household Total<br>Income |
|------------------|------------------------------|-------------------------------------|---------------------------|---------------------------|
| CDE <sub>0</sub> | 0.105<br>(0.066)             | -0.025<br>(0.071)                   | 1.116<br>(1.366)          | 1.067<br>(2.154)          |
| CDE <sub>1</sub> | 0.013<br>(0.083)             | -0.094<br>(0.075)                   | 1.280<br>(1.788)          | 3.884<br>(2.341)          |
| CTE              | 0.099<br>(0.107)             | -0.386<br>(0.121)                   | 3.323<br>(2.383)          | 1.537<br>(3.239)          |
| LATE             | 0.061<br>(0.052)             | -0.113<br>(0.051)                   | 1.412<br>(1.093)          | 2.258<br>(1.569)          |

Estimates for CDE<sub>0</sub>, CDE<sub>1</sub>, CTE (as in (8)), and LATE. Income measured in thousands of dollars per year. Estimates account for the person-level weight for the adult survey in interim analyses. Standard errors are displayed in parentheses.

modification of the sequential ignorability assumption of [Imai et al. \(2010\)](#) that was proposed by [Yamamoto \(2013\)](#) for applications with instrumental variables.

Our results imply EIMC and restriction (7) identify CDE<sub>0</sub>, CDE<sub>1</sub>, and CTE; see Theorem 4.3. More generally, our results imply (9) is identified if and only if

$$E_{Q_0}[\sum_{z \in \mathbf{Z}} 1\{T^*(z) = t\} \nu(z, X) | V^*(t), X] = E_{Q_0}[\ell(T^*, X) | V^*(t), X] \quad (10)$$

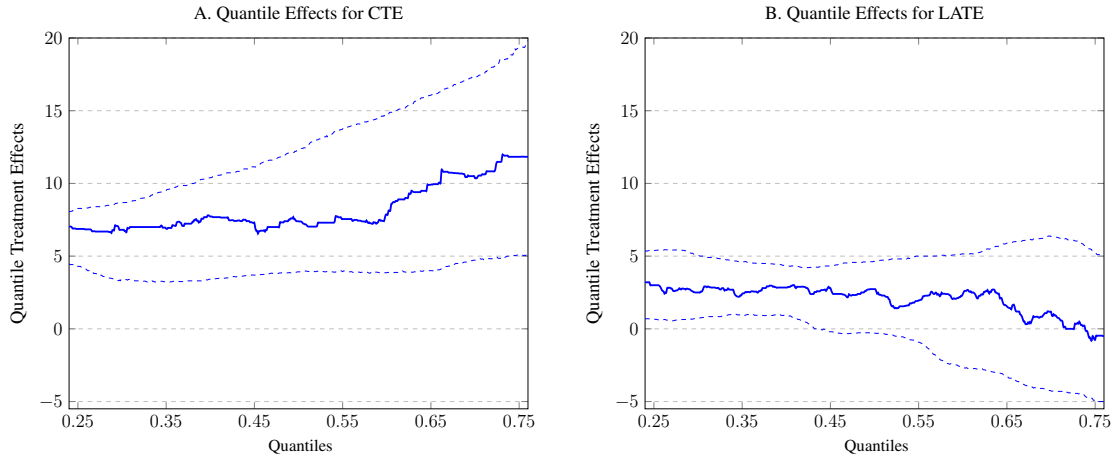
for some function  $\nu$  of  $(Z, X)$  and  $V^*(t)$  a known transformation of  $T^*$ .<sup>8</sup> Setting  $\kappa(z, X) = \nu(z, X)/P(Z = z|X)$  for any  $\nu$  solving (10), our results then imply (9) is identified by

$$E_{Q_0}[\rho(Y^*(t)) \ell(T^*, X)] = E_P[\rho(Y) 1\{T = t\} \kappa(Z, X)]. \quad (11)$$

Table II reports treatment effects estimates based on (11) for different outcomes. LATE estimates reveal a statistically significant effect only on labor force participation. LATE estimates on income measures are both positive, but statistically insignificant. Decomposing these LATE estimates into CDE<sub>0</sub>, CDE<sub>1</sub>, and CTE (as defined in (8)) provides insights into

<sup>8</sup>Specifically, we set  $V^*(t) = 0$  when  $t = (0, 0)$  and  $T^* \in \{\text{CN}, \text{CC}\}$ ,  $V^*(t) = 1$  when  $t = (1, 1)$  and  $T^* \in \{\text{CA}, \text{CC}\}$ , and  $V^*(t) = T^*$  for all other values of  $t$  and  $T^*$ .

FIGURE 1.—Quantile Treatment Effects for Total Household Income



Graph A: Difference in quantiles of  $Y^*(1, 1)$  and  $Y^*(0, 0)$  for CC. Graph B: Difference in the quantiles of  $Y^*(1, M^*(1))$  and  $Y^*(0, M^*(0))$  for  $\{CA, CN, CC\}$ . Income measured in thousands of dollars per year. Estimates account for the person-level weight for the adult survey in interim analyses. Dashed lines are 95% pointwise equitailed bootstrap confidence regions.

the role of mental health as a mediator.  $CDE_0$  and  $CDE_1$ , which report treatment effects for types CN and CA, suggest a positive effect on labor force participation but are statistically insignificant. CTE for labor force participation, which reports treatment effects for type CC, is in contrast statistically significant and an order of magnitude larger than  $CDE_0$  and  $CDE_1$ . Because CC is the only complier type whose mental health changes after relocation, these results lend credence to the conjecture by Ludwig et al. (2008) that improved mental health is a mediator through which relocation affects labor force participation.

While the positive CTE and LATE estimates for income measures are not statistically significant, we find significant Quantile Treatment Effects (QTEs) for type CC on Household Total Income. Figure 1 reports these estimates with 95% (pointwise) bootstrap equitailed confidence regions. QTEs corresponding to both relocating and improving mental health (CTE) are positive and statistically significant across the quantiles we examine. Reflecting the low proportion of type CC, the LATE QTEs exhibit mixed findings, being statistically significant for lower quantiles only.

#### 4. IDENTIFICATION

In this section, we characterize point identification and provide a recipe for estimation.

#### 4.1. Two Key Lemmas

We begin by introducing two lemmas that play a fundamental role in our analysis.

LEMMA 4.1: *If Assumptions 2.1 and 2.2 hold, then  $\Theta_0$  is convex and there is a  $\bar{Q} \in \Theta_0$  such that all  $Q \in \Theta_0$  are absolutely continuous with respect to  $\bar{Q}$ .*

Lemma 4.1 shows there is a distribution  $\bar{Q}$  in the identified set  $\Theta_0$  that is “larger” than any other distribution in  $\Theta_0$ . By “larger” we mean that the support of any distribution in the identified set must be contained in the support of  $\bar{Q}$ . There may be multiple distributions  $\bar{Q}$  satisfying the conclusion of Lemma 4.1. However, such distributions are equivalent in the sense that they must assign zero probability to the same sets.

Our second lemma is the cornerstone of our identification analysis. In order to formally state this key result, we introduce a linear operator  $\Upsilon$  that maps functions of  $(Y, T, Z, X)$  to functions of  $(Y^*, T^*, X)$ . Specifically, for any  $f \in L^1(P)$  we set

$$\Upsilon(f) \equiv \sum_{t \in \mathbf{T}} E_{P_{Z|X}}[f(Y^*(t), t, Z, X) 1\{T^*(Z) = t\}], \quad (12)$$

where  $P_{Z|X}$  denotes the conditional distribution of  $Z$  given  $X$  and the notation  $E_{P_{Z|X}}$  emphasizes the expectation is taken with respect to  $Z$  while  $(Y^*, T^*, X)$  are kept fixed. Under Assumption 2.1, it is possible to show that  $\Upsilon(f)$  in fact satisfies

$$\Upsilon(f) = E_Q[f(Y, T, Z, X) | Y^*, T^*, X] \quad (13)$$

for any  $Q$  in the identified set  $\Theta_0$  – i.e.,  $\Upsilon$  maps functions of observables into functions of unobservables by taking conditional expectations given  $(Y^*, T^*, X)$ .

The next lemma combines the map  $\Upsilon$  with the distribution  $\bar{Q}$  to obtain a sufficient condition for the expectation of  $\ell(Y^*, T^*, X)$  to be identified for a given function  $\ell$ .

LEMMA 4.2: *Let Assumptions 2.1 and 2.2 hold, and  $\ell$  satisfy  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$  for some  $\kappa \in L^1(P)$ . Then it follows that  $\ell \in L^1(Q)$  for all  $Q \in \Theta_0$  and in addition*

$$E_Q[\ell(Y^*, T^*, X)] = E_P[\kappa(Y, T, Z, X)]. \quad (14)$$

Lemma 4.2 follows from noting that the conditions imposed on  $\kappa$  and result (13) imply

$$E_Q[\kappa(Y, T, Z, X) | Y^*, T^*, X] = \ell(Y^*, T^*, X)$$

for any  $Q \in \Theta_0$ , and therefore result (14) is immediate by the law of iterated expectations. The main implication of Lemma 4.2 is a recipe for identification and estimation. In particular, Lemma 4.2 suggests estimating the expectation of  $\ell(Y^*, T^*, X)$  by employing sample moments based on an estimator of a function  $\kappa$  satisfying  $\Upsilon(\kappa) = \ell$ . To this end, it is fruitful to rely on our next corollary, which represents  $\kappa$  in terms of the density  $\pi \equiv dP_{Z|X}/d\mu_{Z|X}$ .

**COROLLARY 4.1:** *Let Assumptions 2.1 and 2.2 hold, and  $\nu \in L^1(P)$  satisfy*

$$\mu\left(\sum_{t \in \mathbf{T}} E_{\mu_{Z|X}}[\nu(Y^*(t), t, Z, X) 1\{T^*(Z) = t\}]\right) = \ell(Y^*, T^*, X) = 1. \quad (15)$$

*If  $\mu(\pi(Z, X) > \delta) = 1$  for some  $\delta > 0$ , then the function  $\kappa \equiv \nu/\pi$  satisfies  $\kappa \in L^1(P)$  and  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$ , and therefore  $E_{Q_0}[\ell(Y^*, T^*, X)] = E_P[\kappa(Y, T, Z, X)]$ .*

Corollary 4.1 shows that we may find the desired  $\kappa$  by taking the ratio of a function  $\nu$  satisfying (15) and the density  $\pi \equiv dP_{Z|X}/d\mu_{Z|X}$ . In the majority of our examples,  $\mu$  can be chosen to be known and  $\nu$  can be computed analytically or numerically. Hence, in these cases,  $\kappa$  is known up to  $\pi$ . Our estimators in Section 5 heavily rely on this observation.

**REMARK 4.1:** Example 2.1, based on Rosenbaum and Rubin (1983), helps illustrate Corollary 4.1. Recall that we set  $\mu(T^*(Z) = Z) = 1$  and  $\ell(Y^*, T^*, X) = Y^*(1) - Y^*(0)$ . Corollary 4.1 suggests identifying the ATE by finding a  $\nu$  satisfying (15). To this end, it is convenient to set  $\mu$  to satisfy  $\mu(Z = 0|X) = \mu(Z = 1|X) = 1/2$ , which implies  $\nu(Y, T, Z, X) = 2Y(1\{Z = 1\} - 1\{Z = 0\})$  solves (15) and the density  $\pi \equiv dP_{Z|X}/d\mu_{Z|X}$  equals  $\pi(z, X) = 2P(Z = z|X)$ . Corollary 4.1 and computing  $\kappa \equiv \nu/\pi$  then yield

$$E_{Q_0}[Y^*(1) - Y^*(0)] = E_P[\kappa(Y, T, Z, X)] = E_P\left[\frac{Y 1\{Z = 1\}}{P(Z = 1|X)} - \frac{Y 1\{Z = 0\}}{P(Z = 0|X)}\right],$$

which recovers the propensity score reweighing moment for identifying the ATE. Similarly, applying Corollary 4.1 to the model in Imbens and Angrist (1994) recovers the  $\kappa$ -weights identifying equations of Abadie (2003). In both cases, the distribution of  $Y^*$  under  $\mu$  can be left unspecified and the distribution of  $X$  under  $\mu$  set to equal the distribution under  $P$ . More generally, in models that only restrict the support of  $T^*$ , it is often computationally convenient to set  $\mu$  so that  $(T^*, Z, X)$  are mutually independent and  $T^*$  and  $Z$  are uniformly distributed. We employ this approach below when revisiting Examples 2.4 and 2.5. ■

## 4.2. Main Result

Lemma 4.2 shows that a sufficient condition for identifying the expectation

$$\lambda_{Q_0} \equiv E_{Q_0}[\ell(Y^*, T^*, X)] \quad (16)$$

is that there be a function  $\kappa$  satisfying  $\Upsilon(\kappa) = \ell$ . Moreover, Lemma 4.2 also suggests an estimation strategy for  $\lambda_{Q_0}$ . We next show this sufficient condition for identification is close to necessary and, as a result, the estimation strategy applies whenever  $\lambda_{Q_0}$  is identified.

Concretely, we will show  $\lambda_{Q_0}$  is identified *if and only if* the function  $\ell$  is in the closure of the set of functions that equal  $\Upsilon(\kappa)$  for some  $\kappa$  – notice the contrast with Lemma 4.2, which requires  $\ell$  to *equal*  $\Upsilon(\kappa)$  for some  $\kappa$ . This result implies that for  $\lambda_{Q_0}$  to be identified,  $\ell$  must be the limit of functions whose expectations are identified through Lemma 4.2. This characterization is constructive in that it suggests estimating  $\lambda_{Q_0}$  by employing the sample moments of a sequence of functions  $\{\kappa_j\}$  for which  $\Upsilon(\kappa_j)$  converges to  $\ell$ . Formalizing this discussion, however, first requires clarifying the sense (i.e. topology) in which we mean converge, closure, and limit. To this end, we introduce one additional assumption.

**ASSUMPTION 4.1:** (i)  $\ell \in L^1(Q)$  for all  $Q \in \Theta_0$ ; (ii) The density  $dQ/d\mu$  belongs to the interior of  $\mathcal{Q}$  in  $\mathbf{Q}$  for some  $Q \in \Theta_0$ .

Assumption 4.1(i) requires  $\ell$  be integrable with respect to every  $Q \in \Theta_0$ . This condition can be ensured by imposing suitable regularity conditions through the choice of  $\mathcal{Q}$  in Assumption 2.2(iii). Assumption 4.1(ii) is needed to show the necessity, but not the sufficiency, of our conditions for point identification of  $\lambda_{Q_0}$ . Intuitively, Assumption 4.1(ii) implies that the restrictions imposed through  $\mathcal{Q}$  do not point identify  $\lambda_{Q_0}$ .

As we discussed, identification of  $\lambda_{Q_0}$  turns on whether  $\ell$  belongs to the closure of the set of functions that equal  $\Upsilon(\kappa)$  for some  $\kappa$ . The latter set is the range of  $\Upsilon$ , denoted by

$$\mathcal{R} \equiv \left\{ f \in \bigcap_{Q \in \Theta_0} L^1(Q) : f = \Upsilon(\kappa) \text{ for some } \kappa \in L^1(P) \right\}.$$

To introduce the topology determining the closure of  $\mathcal{R}$ , we define  $\langle f, g \rangle_Q \equiv \int f g dQ$ , let  $Q_V$  denote the marginal distribution of a random variable  $V$  under  $Q$ , and define the set

$$\mathcal{S}_Q \equiv \left\{ s \in L^\infty(Q_{Y^*T^*X}) \text{ such that } s \frac{dQ}{d\mu} \in \mathbf{Q} \right\},$$

where  $L^\infty(Q_{Y^*T^*X})$  is the set of functions of  $(Y^*, T^*, X)$  that are bounded with probability one under  $Q$ . Additionally, we let  $\text{span}\{A\}$  denote the linear span of a set  $A$ , and introduce a vector space  $\mathcal{B}$  of linear functionals defined on  $\bigcap_{Q \in \Theta_0} L^1(Q)$  by setting

$$\mathcal{B} \equiv \text{span}\{B : \bigcap_{Q \in \Theta_0} L^1(Q) \rightarrow \mathbf{R} \text{ such that } B = \langle \cdot, s \rangle_Q \text{ for some } s \in \mathcal{S}_Q, Q \in \Theta_0\}.$$

Crucially, by Assumption 4.1(i) and Lemma 4.2,  $\ell$  and all functions  $f \in \mathcal{R}$  belong to the domain of all the linear functionals  $B \in \mathcal{B}$ . Using this observation, we define the topology  $\tau$  on  $\mathcal{R} \cup \{\ell\}$  to be the weak topology generated by the functionals  $B \in \mathcal{B}$  – i.e., a sequence  $\{f_j\}$  in  $\mathcal{R}$  converges to  $\ell$  under  $\tau$  if and only if  $B(f_j) \rightarrow B(\ell)$  for all  $B \in \mathcal{B}$ .

The next theorem is our main identification result.

**THEOREM 4.1:** *If Assumptions 2.1, 2.2, and 4.1 hold, then it follows that  $\lambda_{Q_0}$  is identified if and only if  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$ .*

Intuitively, Theorem 4.1 shows  $\lambda_{Q_0}$  is identified if and only if there is a sequence  $\{f_j\} \subseteq \mathcal{R}$  converging to  $\ell$  under  $\tau$ . To gain insight into why this condition is sufficient for identification, let  $\mathbf{1}$  be the function that is constant at one, set  $B_{Q_0} \equiv \langle \cdot, \mathbf{1} \rangle_{Q_0}$ , and note

$$\lambda_{Q_0} = \langle \ell, \mathbf{1} \rangle_{Q_0} = B_{Q_0}(\ell). \quad (17)$$

Since  $f_j \in \mathcal{R}$  implies  $f_j = \Upsilon(\kappa_j)$  for some function  $\kappa_j$  of  $(Y, T, Z, X)$ , it follows from  $\{f_j\}$  converging to  $\ell$  in the  $\tau$  topology that there is a sequence  $\{\kappa_j\}$  satisfying

$$B_{Q_0}(\ell) = \lim_{j \rightarrow \infty} B_{Q_0}(f_j) = \lim_{j \rightarrow \infty} \langle \Upsilon(\kappa_j), \mathbf{1} \rangle_{Q_0} = \lim_{j \rightarrow \infty} E_P[\kappa_j(Y, T, Z, X)], \quad (18)$$

where the final equality follows from Lemma 4.2. Results (17) and (18) not only establish the identification of  $\lambda_{Q_0}$ , but also suggest an estimation strategy: Simply employ sample moments based on estimates of the functions  $\{\kappa_j\}$ . As illustrated by Remark 4.1, the sequence  $\{\kappa_j\}$  may be intuitively understood as a generalization of propensity score reweighting or Abadie's kappa to the large class of causal models that we study.

The conclusion that the existence of the desired  $\{\kappa_j\}$  is necessary for the identification of  $\lambda_{Q_0}$  is more surprising.<sup>9</sup> Importantly, necessity implies it is without loss of generality to estimate  $\lambda_{Q_0}$  by employing an estimation strategy based on (18). The next corollary illustrates how Theorem 4.1 can also characterize  $\{\kappa_j\}$ , which is helpful for estimation.

**COROLLARY 4.2:** *Let Assumptions 2.1, 2.2 hold with  $\mathcal{Q} = \mathbf{Q} = L^\infty(\mu)$ ,  $\mu \ll \bar{Q}$  with  $d\mu/d\bar{Q}$  bounded, and  $\ell \in L^1(\mu_{Y^*T^*X})$ . Then, it follows that:*

- (i)  $\lambda_{Q_0}$  is identified if and only if  $\lim_{j \rightarrow \infty} \|\ell - \Upsilon(\kappa_j)\|_{\mu,1} = 0$  for some  $\{\kappa_j\} \subseteq L^1(P)$ .  
Moreover, any such sequence  $\{\kappa_j\}$  satisfies  $\lambda_{Q_0} = \lim_{j \rightarrow \infty} E_P[\kappa_j(Y, T, Z, X)]$ .
- (ii) Suppose in addition that  $\mu(\pi(Z, X) > \delta) = 1$  for some  $\delta > 0$  and  $\pi = dP_{Z|X}/d\mu_{Z|X}$ .  
Then,  $\lambda_{Q_0}$  is identified if and only if there is a sequence  $\{\nu_j\} \subseteq L^1(P)$  satisfying

$$\lim_{j \rightarrow \infty} E_\mu[\ell(Y^*, T^*, X) - \sum_{t \in \mathbf{T}} E_{\mu_{Z|X}}[\nu_j(Y^*(t), t, Z, X) 1\{T^*(Z) = t\}]] = 0. \quad (19)$$

Moreover, for any such  $\{\nu_j\}$ ,  $\kappa_j \equiv \nu_j/\pi$  satisfies  $\lambda_{Q_0} = \lim_{j \rightarrow \infty} E_P[\kappa_j(Y, T, Z, X)]$ .

Corollary 4.2 specializes Theorem 4.1 by imposing that Assumption 2.2(iii) only require the density of  $Q_0$  to be bounded. The corollary also assumes  $\mu \ll \bar{Q}$ , which ensures that every response type allowed for by  $\mu$  has positive probability under  $\bar{Q}$ .<sup>10</sup> Corollary 4.2(i) shows  $\lambda_{Q_0}$  is identified if and only if  $\ell$  is in the closure of  $\mathcal{R}$  under  $\|\cdot\|_{\mu,1}$ . Corollary 4.2(ii) provides conditions under which we can set  $\kappa_j = \nu_j/\pi$  for any  $\{\nu_j\}$  satisfying (19) and  $\pi \equiv dP_{Z|X}/d\mu_{Z|X}$ . This result has two important implications. First, Corollary 4.2(ii) provides a characterization of  $\{\kappa_j\}$  that we use in estimation. Second, it allows us to assess whether the restrictions of our model point identify  $\lambda_{Q_0}$ ; see Example 2.5 below.

**REMARK 4.2:** Our main identification results do not fundamentally rely on  $T$  being discrete. Under a mild modification to the proofs, Theorem 4.1 and Corollary 4.2(i) hold when  $T$  is continuous provided that for any  $f(Y, T, Z, X)$  we understand  $\Upsilon(f)$  to be

$$\Upsilon(f) \equiv E_{P_{Z|X}}[f(Y^*(T^*(Z)), T^*(Z), Z, X)]. \quad (20)$$

<sup>9</sup>Theorem 4.1 only implies the existence of a net, which is a generalized sequence whose index set is a possibly uncountable directed set, rather than the natural numbers. However, we discuss sequences for ease of exposition.

<sup>10</sup>E.g., in Imbens and Angrist (1994), the requirement  $\mu \ll \bar{Q}$  rules out compliers having probability one.

Note that this definition for  $\Upsilon$  coincides with the one in (12) when  $T$  is discrete, but it remains valid when  $T$  is continuous as well. Some of our forthcoming results, such as Theorem 4.3 below, rely more essentially on  $T$  being discrete. ■

### 4.3. Special Case: Types

Features of the distribution of  $(T^*, X)$  are identified by specializing our analysis to expectations of functions that depend on  $(T^*, X)$  only – i.e., to parameters with the structure

$$\lambda_{Q_0} \equiv E_{Q_0}[\ell(T^*, X)]. \quad (21)$$

While Theorem 4.1 applies to these parameters, the restriction that  $\ell$  depend only on  $(T^*, X)$  enables sharper conclusions. In particular, we will show  $\lambda_{Q_0}$  is identified if and only if it is identified from the distribution of  $(T, Z, X)$ . To this end, we define

$$\bar{Q}^{\text{it}} \equiv \bar{Q}_{Y^*|X} \bar{Q}_{T^*|ZX},$$

which shares the same marginal distributions for  $(Y^*, X)$  and  $(T^*, X)$  as  $\bar{Q}$ , but is such that  $Y^*$  is independent of  $T^*$  conditionally on  $X$ . Given this notation we impose:

**ASSUMPTION 4.2:** (i)  $\bar{Q}^{\text{it}} \ll \bar{Q}$ ; (ii)  $(d\bar{Q}_{Y^*T^*X}^{\text{it}}/d\bar{Q}_{Y^*T^*X})s \in \mathcal{S}_{\bar{Q}}$  for all  $s \in \mathcal{S}_{\bar{Q}} \cap L^\infty(\bar{Q}_{T^*X})$ ; (iii)  $(dQ_{T^*X}/d\bar{Q}_{T^*X})s \in \mathcal{S}_{\bar{Q}}$  for all  $Q \in \Theta_0$  and  $s \in L^\infty(Q_{T^*X}) \cap \mathcal{S}_Q$ ; (iv) For any  $Q \in \Theta_0$  and  $s \in \mathcal{S}_Q$  we have that  $E_Q[s(Y^*, T^*, X)|T^*, X] \in \mathcal{S}_Q$ .

Assumption 4.2(i) requires the support of  $Y^*$  conditional on  $(T^*, X)$  under  $\bar{Q}$  to not depend on  $T^*$ . Assumptions 4.2(ii) and 4.2(iii) impose restrictions on the densities  $d\bar{Q}^{\text{it}}/d\bar{Q}$  and  $dQ_{T^*X}/d\bar{Q}_{T^*X}$  (for  $Q \in \Theta_0$ ), while Assumption 4.2(iv) requires that conditional expectations of functions in  $\mathcal{S}_Q$  belong to  $\mathcal{S}_Q$  as well. Assumptions 4.2(ii)-(iv) can in many applications be verified by appropriately selecting  $\mathcal{Q}$  and  $\mathbf{Q}$  in Assumption 2.2(iii).

The next result characterizes identification of features of the distribution of  $(T^*, X)$ .

**THEOREM 4.2:** Let Assumptions 2.1, 2.2, 4.1, 4.2 hold,  $\lambda_{Q_0}$  be as in (21), and define

$$\mathcal{R}_T \equiv \left\{ f \in \bigcap_{Q \in \Theta_0} L^1(Q) : f = \Upsilon(\kappa) \text{ for some } \kappa \in L^1(P_{TZ X}) \right\}.$$

Then, it follows that  $\lambda_{Q_0}$  is identified if and only if  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}_T$ .

Theorem 4.2 implies that identification of  $\lambda_{Q_0}$  turns on whether  $\ell$  can be approximated by a sequence  $\{\Upsilon(\kappa_j)\}$  with  $\{\kappa_j\}$  depending only on  $(T, Z, X)$  – a conclusion that contrasts with Theorem 4.1, in which  $\kappa_j$  depends on  $(Y, T, Z, X)$ . When such an approximating sequence exists,  $\lambda_{Q_0}$  equals the limit of the expectations of the corresponding  $\kappa_j$ .

The next result highlights the implications of Theorem 4.2 when instruments are discrete.

**COROLLARY 4.3:** *Let Assumption 2.1, 2.2 hold,  $\mathcal{Q} = \mathbf{Q} = L^1(\mu)$ ,  $\bar{Q}^{\text{it}} \ll \bar{Q}$  with  $d\bar{Q}^{\text{it}}/d\bar{Q}$  bounded,  $\ell \in L^1(\bar{Q}_{T^*X})$ ,  $Z$  be discrete,  $P(Z = z|X) \geq \varepsilon > 0$  a.s. for all  $z \in \mathbf{Z}$ , and  $\bar{Q}(T^* = t^*|X) \geq \varepsilon > 0$  a.s. for any  $t^* \in \mathbf{T}^*$  with  $\bar{Q}(T^* = t^*) > 0$ . Then:*

- (i)  $\lambda_{Q_0} \equiv E_{Q_0}[\ell(T^*, X)]$  is identified if and only if  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$  for some  $\kappa \in L^1(P_{TZ X})$ . Moreover, any such  $\kappa$  satisfies  $\lambda_{Q_0} = E_P[\kappa(T, Z, X)]$ .
- (ii) Suppose in addition that  $\mu \ll \bar{Q}$  and  $\mu(\pi(Z, X) > \delta) = 1$  for some  $\delta > 0$  and  $\pi = dP_{Z|X}/d\mu_{Z|X}$ . Then  $\lambda_{Q_0}$  is identified if and only if there is a  $\nu \in L^1(P_{TZ X})$  satisfying

$$\mu(\ell(T^*, X) = \sum_{t \in \mathbf{T}} E_{\mu_{Z|X}}[\nu(t, Z, X) 1\{T^*(Z) = t\}]) = 1. \quad (22)$$

Moreover, for any such function  $\nu$ ,  $\kappa = \nu/\pi$  satisfies  $\lambda_{Q_0} = E_P[\kappa(T, Z, X)]$ .

Corollary 4.3(i) specializes Theorem 4.2 to the case of discrete instruments and no identifying assumptions being imposed through  $\mathcal{Q}$ . In this context, Corollary 4.3(i) provides a converse to Lemma 4.2 by showing  $\lambda_{Q_0}$  is identified if and only if  $\ell$  equals  $\Upsilon(\kappa)$  for some function  $\kappa$  of  $(T, Z, X)$ . Corollary 4.3(ii) parallels Corollary 4.2 in providing conditions that are useful for assessing whether  $\lambda_{Q_0}$  is identified and estimating it if it is. Together with Corollary 4.4 below, Corollary 4.3(ii) provides conditions under which the sufficient condition for identification derived by Heckman and Pinto (2018) for discrete instrument applications (such as Example 2.4) is also necessary.<sup>11</sup>

#### 4.3.1. Examples Revisited

We illustrate our results by revisiting Examples 2.4 and 2.5.

<sup>11</sup>Specifically, in a setting with no covariates  $\ell$  may be interpreted as a vector. Theorem T-2 in Heckman and Pinto (2018) then shows  $E_{Q_0}[\ell(T^*)]$  is identified when  $\ell$  is orthogonal to the null space of a matrix  $B$ . In contrast, in this context (22) may be understood as the equivalent condition that  $\ell$  belong to the range of the adjoint  $B'$ .

**Example 2.4 (cont.)** In this example,  $Z$  and  $T^*$  were discrete and  $\mu$  imposed  $\mu(T^* \in \mathbf{R}^*) = 1$  for a set  $\mathbf{R}^* \equiv \{t_1^*, \dots, t_r^*\}$ . Let  $\mathbf{Z} \equiv \{z_1, \dots, z_q\}$  denote the support of  $Z$  and  $\omega_j(t^*) \equiv (1\{t^*(z_j) = t_1\}, \dots, 1\{t^*(z_j) = t_d\})'$ . As in Remark 4.1, we pick  $\mu$  with  $Z$  independent of  $X$  under  $\mu$ . Corollary 4.3(ii) implies the expectation of  $\ell(T^*, X)$  is identified if and only if

$$\min_{\{s_j\}_{j=1}^q \subset \mathbf{R}^d} \sum_{i=1}^r (\ell(t_i^*, X) - \sum_{j=1}^q s_j' \omega_j(t_i^*))^2 = 0 \quad (23)$$

with probability one (over  $X$ ). Moreover, provided (23) holds, we can find a function  $\kappa$  of  $(T, Z, X)$  whose expectation equals the expectation of  $\ell(T^*, X)$  by setting  $(\kappa(t_1, z_j, X), \dots, \kappa(t_d, z_j, X)) \equiv (s_j(X)/P(Z = z_j|X))'$  for any  $(s_1(X), \dots, s_q(X))$  minimizing (23). Specializing (23) to Kline and Walters (2016) implies the distribution of  $(T^*, X)$  is identified in that context. More generally, we highlight that identifying features of the distribution of  $(T^*, X)$  reduces to a least squares problem when  $Z$  is discrete. ■

**Example 2.5 (cont.)** In this example,  $Z = (C, W)$  with  $C \in \{0, 1\}$ ,  $W \in \mathbf{R}$ , and  $\mu$  required that  $T^*(c, w)$  be increasing in  $c$  and decreasing in  $w$ . Assuming  $W$  is continuously distributed with compact support  $[\underline{w}, \bar{w}]$ , it is possible to show there are  $(K_0^*, K_1^*)$  with

$$T^*(c, w) = \sum_{i=0}^1 1\{c = i, K_i^* > w\} \quad (24)$$

and  $K_i^* \in [\underline{w}, \bar{w}] \cup \{\infty\}$ . We study identification of the distribution of  $(K_0^*, K_1^*, X)$  and note the restriction that  $T^*(c, w)$  be increasing in  $c$  is equivalent to  $\mu(K_1^* \geq K_0^*) = 1$ . It is convenient to let  $K_i^*$  be continuously distributed on  $(\underline{w}, \bar{w}]$  under  $\mu$ , but allow  $\mu$  to assign positive mass to  $\{\underline{w}\}$  and  $\{\infty\}$  – i.e. to assign positive probability to always takers and never takers. Under conditions paralleling those in Corollary 4.2(ii), Theorem 4.2 then implies that the expectation of a function  $\ell$  of  $(K_0^*, K_1^*, X)$  is identified if and only if

$$\lim_{j \rightarrow \infty} E_\mu \left[ \sum_{i=0}^1 \int_{K_i^*}^{\bar{w}} \nu_j(0, i, w, X) dw + \int_{\underline{w}}^{K_i^*} \nu_j(1, i, w, X) dw - \ell(K_0^*, K_1^*, X) \right] = 0 \quad (25)$$

for some sequence  $\{\nu_j\}$  of functions of  $(T, C, W, X)$ . Moreover, provided condition (25) holds, the expectation of  $\ell(K_0^*, K_1^*, X)$  equals the limit of the expectations of the functions

$$\kappa_j(t, c, w, X) \equiv \frac{\nu_j(t, c, w, X)}{f_{W|CX}(w|c, X)P(C = c|X)}$$

where  $f_{W|CX}$  denotes the density of  $W$  given  $(C, X)$ . More generally, Theorem 4.2 implies the expectation of  $\ell(K_0^*, K_1^*, X)$  is identified if and only if  $\ell$  belongs to the closure of  $\mathcal{T} \equiv \{f : f(K_0^*, K_1^*, X) = g_0(K_0^*, X) + g_1(K_1^*, X) \text{ for some } g_0, g_1\}$  under  $\|\cdot\|_{\mu,1}$ . For instance, the probability of the event  $\{K_1^* > a_1, K_0^* \leq a_0\}$  is identified if and only if  $a_0 \geq a_1$ . ■

#### 4.4. Special Case: Outcomes

We conclude our identification analysis by focusing on parameters with the structure

$$\lambda_{Q_0} \equiv E_{Q_0}[\rho(Y^*(t), X)\ell(T^*, X)], \quad (26)$$

for some  $t \in \mathbf{T}$  and functions  $\rho$  and  $\ell$ . These parameters include features of the conditional distribution of a potential outcome given types and covariates as a special case. Intuitively, identification of these parameters can only be possible from the distribution of observations for which treatment assignment  $T$  equals  $t$ . To formalize this intuition, we first define

$$\bar{Q}^{\text{io}} \equiv \bar{Q}_{Y^*(t_1)|X} \cdots \bar{Q}_{Y^*(t_d)|X} \bar{Q}_{T^*ZX};$$

i.e.,  $\bar{Q}^{\text{io}}$  has the same marginal distributions for  $(Y^*(t), X)$  and  $(T^*, X)$  as  $\bar{Q}$ , but is such that all coordinates of  $Y^*$  and  $T^*$  are mutually conditionally independent. We also define

$$\phi_{\bar{Q},\rho}(Y^*(t), X) \equiv \frac{\rho(Y^*(t), X) - E_{\bar{Q}}[\rho(Y^*(t), X)|X]}{\text{Var}_{\bar{Q}}\{\rho(Y^*(t), X)|X\}},$$

where we set  $\phi_{\bar{Q},\rho}(Y^*(t), X) = 0$  when  $\text{Var}_{\bar{Q}}\{\rho(Y^*(t), X)|X\} = 0$ .

Given the introduced notation, we impose the following assumptions.

ASSUMPTION 4.3:  $\rho \in L^\infty(Q)$  and  $\ell \in L^1(Q)$  for all  $Q \in \Theta_0$ .

ASSUMPTION 4.4: (i)  $E_Q[\rho(Y^*(t), X)|T^*, X] \in \mathcal{S}_Q$  and  $E_Q[s(Y^*, T^*, X)|T^*, X] \in \mathcal{S}_Q$  for any  $Q \in \Theta_0$  and  $s \in \mathcal{S}_Q$ ; (ii)  $\bar{Q}^{\text{io}} \ll \bar{Q}$  and  $d\bar{Q}^{\text{io}}/d\bar{Q} \in L^\infty(\bar{Q})$ ; (iii)  $\phi_{\bar{Q},\rho} \in L^\infty(\bar{Q})$ ,  $\text{Var}_{\bar{Q}}\{\rho(Y^*(t), X)|X\} > 0$  a.s. under  $\bar{Q}$ , and  $s(d\bar{Q}_{Y^*T^*X}^{\text{io}}/d\bar{Q}_{Y^*T^*X})\phi_{\bar{Q},\rho} \in \mathcal{S}_{\bar{Q}}$  for all  $s \in L^\infty(\bar{Q}_{T^*X}) \cap \mathcal{S}_{\bar{Q}}$ ; (iv)  $(dQ_{T^*X}/d\bar{Q}_{T^*X})s \in \mathcal{S}_{\bar{Q}}$  for all  $Q \in \Theta_0$  and  $s \in L^\infty(Q_{T^*X}) \cap \mathcal{S}_Q$ .

Assumption 4.3 ensures  $\ell\rho$  is integrable for all  $Q \in \Theta_0$  and therefore that  $\lambda_{Q_0}$  is well defined. Assumption 4.4 is similar in spirit to the conditions we imposed in Assumption 4.2 for studying features of the distribution of  $(T^*, X)$ . Specifically, Assumptions 4.4(i)(iv)

parallel Assumptions 4.2(ii)-(iv), while Assumption 4.4(ii) imposes a key support requirement that parallels Assumption 4.2(i). Finally, Assumption 4.4(iii) ensures that  $\rho$  is not constant, and that therefore (26) does not fall within the framework of Section 4.3.

The next result shows parameters with the structure in (26) are identified if and only if they are identified from the distribution of observations with treatment  $T$  equal to  $t$ .

**THEOREM 4.3:** *Let Assumptions 2.1, 2.2, 4.1(ii), 4.3, 4.4 hold,  $\lambda_{Q_0}$  be as in (26), define  $L^1(P_{tZX}) \equiv \{f \in L^1(P) : f(T, Z, X) = 1\{T = t\}g(Z, X) \text{ for some } g \in L^1(P)\}$  and*

$$\mathcal{R}_t \equiv \{f \in \bigcap_{Q \in \Theta_0} L^1(Q) : f = \Upsilon(\kappa) \text{ for some } \kappa \in L^1(P_{tZX})\}.$$

*Then, it follows that  $\lambda_{Q_0}$  is identified if and only if  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}_t$ .*

We emphasize the contrast with Theorem 4.2, which showed identification of the expectation of  $\ell(T^*, X)$  is equivalent to  $\ell$  being in the  $\tau$ -closure of  $\mathcal{R}_T$  (instead of  $\mathcal{R}_t$  in Theorem 4.3). In particular, since  $\mathcal{R}_t \subseteq \mathcal{R}_T$ , it follows that identification of the expectation of  $\rho(Y^*(t), X)\ell(T^*, X)$  implies the identification of the expectation of  $\ell(T^*, X)$ . More generally, since  $\ell$  belonging to  $\mathcal{R}_t$  does not depend on the choice of  $\rho$ , the identification of  $\lambda_{Q_0}$  for some  $\rho$  implies that  $\lambda_{Q_0}$  is in fact identified for all suitable  $\rho$ . Our next corollary illustrates these implications in the case of discrete instruments.

**COROLLARY 4.4:** *Let Assumptions 2.1, 2.2 hold,  $\mathcal{Q} = \mathbf{Q} = L^1(\mu)$ ,  $\bar{Q}^{\text{io}} \ll \bar{Q}$  with  $d\bar{Q}^{\text{io}}/d\bar{Q}$  and  $\rho$  bounded,  $\ell \in L^1(\bar{Q}_{T^*X})$ , and  $\text{Var}_{\bar{Q}}\{\rho(Y^*(t), X)|X\} \geq \varepsilon > 0$  a.s.. If  $Z$  is discrete,  $P(Z = z|X) \geq \varepsilon > 0$  a.s. for all  $z \in \mathbf{Z}$ , and  $\bar{Q}(T^* = t^*|X) \geq \varepsilon > 0$  a.s. for any  $t^* \in \mathbf{T}^*$  with  $\bar{Q}(T^* = t^*) > 0$ , then the following are equivalent:*

- (i)  $E_{Q_0}[\rho(Y^*(t), X)\ell(T^*, X)]$  is identified.
- (ii)  $E_{Q_0}[f(Y^*(t), X)\ell(T^*, X)]$  is identified for any bounded  $f$ .
- (iii)  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$  for some  $\kappa(T, Z, X) = 1\{T = t\}g(Z, X)$  with  $g \in L^1(P_{ZX})$ , and therefore  $E_{Q_0}[f(Y^*(t), X)\ell(T^*, X)] = E_P[f(Y, X)\kappa(T, Z, X)]$  for any bounded  $f$ .

Through the equivalence of (i) and (ii), Corollary 4.4 formalizes that identification of  $\lambda_{Q_0}$  for some  $\rho$  implies identification of  $\lambda_{Q_0}$  for all  $\rho$ . Corollary 4.4 also shows identification of  $\lambda_{Q_0}$  requires that  $\ell = \Upsilon(\kappa)$  for some  $\kappa$  that only employs observations with treatment status  $t$ . Moreover, paralleling Corollaries 4.2 and 4.3, it is possible to show that we may

set  $\kappa(T, Z, X) = 1\{T = t\}\nu(Z, X)/\pi(Z, X)$  with  $\pi = dP_{Z|X}/d\mu_{Z|X}$  and any  $\nu$  satisfying

$$\ell(T^*, X) = E_{\mu_{Z|X}}[\nu(Z, X)1\{T^*(Z) = t\}]. \quad (27)$$

**REMARK 4.3:** Our results also imply necessary and sufficient conditions for the identification of parameters that, for some known  $\rho$  and  $\ell$ , have the structure

$$\frac{E_{Q_0}[\rho(Y^*(t), X)\ell(T^*, X)]}{E_{Q_0}[\ell(T^*, X)]}. \quad (28)$$

These parameters include, for instance, the expectation of  $Y^*(t)$  conditional on  $T^*$  belonging to a set. By Theorems 4.2 and 4.3, if  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}_t$ , then the numerator and denominator in (28) are identified, and therefore so is the ratio. Conversely, if the ratio in (28) is identified to equal some value,  $c_0$ , then  $E_{Q_0}[(\rho(Y^*(t), X) - c_0)\ell(T^*, X)]$  is identified and, by Theorem 4.3,  $\ell$  must belong to the  $\tau$ -closure of  $\mathcal{R}_t$ . Hence, under suitable conditions, identification of (28) is equivalent to  $\ell$  belonging to the  $\tau$ -closure of  $\mathcal{R}_t$ . These observations complement the independent work of Goff (2025), which derives necessary and sufficient conditions for identification of treatment effects conditional on response type in the discrete instrument models studied by Heckman and Pinto (2018). ■

#### 4.4.1. Examples Revisited

**Example 2.4 (cont.)** In this context, Corollary 4.4 and the subsequent discussion imply that the expectation of  $\rho(Y^*(t), X)\ell(T^*, X)$  is identified if and only if

$$\min_{\{s_j\}_{j=1}^q \subset \mathbf{R}} \sum_{i=1}^r (\ell(t_i^*, X) - \sum_{j=1}^q s_j 1\{t_i^*(z_j) = t\})^2 = 0 \quad (29)$$

with probability one. If (29) holds, then the expectation of  $\rho(Y^*(t), X)\ell(T^*, X)$  equals the expectation of  $\rho(Y, X)\kappa(T, Z, X)$  with  $\kappa(T, z_j, X) \equiv 1\{T = t\}s_j(X)/P(Z = z_j|X)$  for any  $(s_1(X), \dots, s_q(X))$  minimizing (29). These results highlight that identifying a functional about outcomes reduces to a least squares problem when  $Z$  is discrete. ■

**Example 2.5 (cont.)** In this application, Theorem 4.3 implies that the expectation of  $\rho(Y^*(0), X)\ell(K_0^*, K_1^*, X)$  is identified if and only if, for some sequence  $\{\nu_j(C, W, X)\}$ ,

$$\lim_{j \rightarrow \infty} E_\mu[\ell(K_0^*, K_1^*, X) - \sum_{c=0}^1 \int_{K_c^*}^{\bar{w}} \nu_j(c, w, X) dw] = 0. \quad (30)$$

Moreover, provided condition (30) holds, the expectation of  $\rho(Y^*(0), X)\ell(K_0^*, K_1^*, X)$  is identified as the limit of the expectations of  $\rho(Y, X)\kappa_j(T, C, W, X)$  with

$$\kappa_j(t, c, w, X) \equiv \frac{1\{t=0\}\nu_j(c, w, X)}{f_{W|CX}(w|c, X)P(C=c|X)}.$$

More generally, our analysis yields that the expectation of  $\rho(Y^*(t), X)\ell(K_0^*, K_1^*, X)$  is identified if and only if  $\ell$  belongs to the  $\|\cdot\|_{\mu,1}$ -closure of the set  $\mathcal{T}_t$ , where

$$\mathcal{T}_0 \equiv \{f : f(K_0^*, K_1^*, X) = g_0(K_0^*, X) + g_1(K_1^*, X) \text{ with } g_0(\infty, X) = g_1(\infty, X) = 0\}$$

$$\mathcal{T}_1 \equiv \{f : f(K_0^*, K_1^*, X) = g_0(K_0^*, X) + g_1(K_1^*, X) \text{ with } g_0(\underline{w}, X) = g_1(\underline{w}, X) = 0\}.$$

For instance, because  $\mu$  satisfies  $\mu(K_1^* \geq K_0) = 1$ , it is possible to show that the parameters

$$E_{Q_0}[Y^*(1) - Y^*(0)|K_0^* \leq a_0, K_1 = a_1] \text{ and } E_{Q_0}[Y^*(1) - Y^*(0)|K_0^* = a_0, K_1 > a_1] \quad (31)$$

are identified if and only if the points  $a_0, a_1$  satisfy  $\underline{w} \leq a_1 \leq a_0 < \infty$ . ■

## 5. ESTIMATION

We have so far allowed features of our model (e.g.,  $\mu$  and  $\ell$ ) to depend on  $P$ . To construct an estimator, however, we need to specify how these features depend on  $P$ . We focus on the leading case in which  $\mu$  and  $\ell$  are known. The parameter of interest remains

$$\lambda_{Q_0} \equiv E_{Q_0}[\ell(Y^*, T^*, X)]. \quad (32)$$

Our estimation strategy is based on two observations that follow from our identification results. First, if the parameter of interest is identified, then it must satisfy

$$\lambda_{Q_0} = \lim_{j \rightarrow \infty} E_P[\kappa_j(Y, T, Z, X)] \quad (33)$$

for some sequence  $\{\kappa_j\}$ . Second,  $\kappa_j$  can often be set to equal  $\kappa_j = \nu_j/\pi$  for some known  $\nu_j$  and  $\pi = dP_{Z|X}/d\mu_{Z|X}$ ; see, e.g., Corollaries 4.1, 4.2, 4.3, and 4.4. The functions  $\{\nu_j\}$  are the only feature of our estimation procedure that is model specific.

We follow the literature in deriving a double robust moment condition. To this end, note

$$E_P\left[\frac{f(Z, X)}{\pi(Z, X)}\right] = E_{P_X}[E_{\mu_{Z|X}}[f(Z, X)]]$$

for any function  $f$  by definition of  $\pi$ . Therefore, the expectation of  $\kappa_j = \nu_j/\pi$  satisfies

$$E_P[\kappa_j(Y, T, Z, X)] = E_P\left[\frac{(\nu_j(Y, T, Z, X) - E_P[\nu_j(Y, T, Z, X)|Z, X])}{\pi(Z, X)}\right] + E_{P_X}[E_{\mu_{Z|X}}[E_P[\nu_j(Y, T, Z, X)|Z, X]]]. \quad (34)$$

This moment is double robust, in the sense that the equality still holds if we substitute  $\pi$  or the conditional expectation of  $\nu_j$  with different functions of  $(Z, X)$ . Double robustness and sample splitting enable estimation through plug-in machine learning methods (Bickel et al., 1993). For concreteness, we follow Smucler et al. (2019) and Chernozhukov et al. (2022a,b) and employ  $\ell_1$ -regularized estimators in the following algorithm:

STEP 1. Partition  $\{1, \dots, n\}$  into  $K$  subsets  $\{I_k\}_{k=1}^K$ . Select functions  $\{b_l\}_{l=1}^p$  of  $(Z, X)$  with  $p$  potentially larger than  $n$ , and let  $b(Z, X) \equiv (b_1(Z, X), \dots, b_p(Z, X))'$ . The number of partitions  $K$  is fixed with  $n$ , and usually set to five or ten.

STEP 2. For each treatment value  $t \in \mathbf{T}$  and partition  $1 \leq k \leq K$  compute the estimators

$$\hat{\beta}_k \in \arg \min_{\tilde{\beta} \in \mathbf{R}^p} \sum_{i \in I_k^c} (\nu_j(Y_i, T_i, Z_i, X_i) - b(Z_i, X_i)' \tilde{\beta})^2 + \alpha \|\tilde{\beta}\|_1 \quad (35)$$

$$\hat{\gamma}_k \in \arg \min_{\tilde{\gamma} \in \mathbf{R}^p} \sum_{i \in I_k^c} \frac{1}{2} (b(Z_i, X_i)' \tilde{\gamma})^2 - E_{\mu_{Z|X}}[b(Z, X_i)' \tilde{\gamma}] + \alpha \|\tilde{\gamma}\|_1, \quad (36)$$

where  $I_k^c = \{1, \dots, n\} \setminus I_k$ . The (possibly data-driven) penalty  $\alpha$  need not be the same in both optimization problems, but the set of functions  $\{b_l\}_{l=1}^p$  should be the same.

STEP 3. For each  $k$ , let  $|I_k|$  denote the number of observations in  $I_k$  and set  $\hat{\lambda}_k$  to equal

$$\hat{\lambda}_k \equiv \frac{1}{|I_k|} \sum_{i \in I_k} \{b(Z_i, X_i)' \hat{\gamma}_k (\nu_j(Y_i, T_i, Z_i, X_i) - b(Z_i, X_i)' \hat{\beta}_k) + E_{\mu_{Z|X}}[b(Z, X_i)' \hat{\beta}_k]\}.$$

STEP 4. The estimator for  $\lambda_{Q_0}$  is given by  $\hat{\lambda} \equiv \sum_k \hat{\lambda}_k \times (|I_k|/n)$ .

We may view  $b(Z, X)' \hat{\beta}_k$  and  $b(Z, X)' \hat{\gamma}_k$  as estimators for the conditional expectation of  $\nu_j$  and  $1/\pi$ , and  $\hat{\lambda}$  as a plug-in estimator based on (34). Sample splitting enables us to relax our assumptions, though we note  $\hat{\lambda}$  is asymptotically normally distributed without sample splitting provided we impose sufficiently strong sparsity requirements. We also observe that in Step 3 we may substitute  $b(Z, X)' \hat{\beta}_k$  and  $b(Z, X)' \hat{\gamma}_k$  with other suitably convergent machine learning estimator (Chernozhukov et al., 2018).

In order to formally study the asymptotic properties of our estimator, we first define

$$\begin{aligned} \beta &\in \arg \min_{\tilde{\beta} \in \mathbf{R}^p} E_P[(\nu_j(Y, T, Z, X) - b(Z, X)' \tilde{\beta})^2] \\ \gamma &\in \arg \min_{\tilde{\gamma} \in \mathbf{R}^p} \left\{ \frac{1}{2} E_P[(b(Z, X)' \tilde{\gamma})^2] - E_{P_X}[E_{\mu_{Z|X}}[b(Z, X)' \tilde{\gamma}]] \right\}, \end{aligned}$$

which are the estimands for which  $\hat{\beta}_k$  and  $\hat{\gamma}_k$  will be assumed to be consistent, and set

$$r_\beta \equiv \max_{1 \leq k \leq K} \{E_P[(b(Z, X)'(\hat{\beta}_k - \beta))^2]\}^{1/2} \quad r_\gamma \equiv \max_{1 \leq k \leq K} \{E_P[(b(Z, X)'(\hat{\gamma}_k - \gamma))^2]\}^{1/2}.$$

The estimands  $b(Z, X)' \beta$  and  $b(Z, X)' \gamma$  are approximations to the conditional expectations of  $\nu_j$  and  $1/\pi$ , and we denote their approximation errors by

$$\begin{aligned} \delta_\beta &\equiv \{E_P[(E_P[\nu_j(Y, T, Z, X)|Z, X] - b(Z, X)' \beta)^2]\}^{1/2} \\ \delta_\gamma &\equiv \{E_P[(\frac{1}{\pi(Z, X)} - b(Z, X)' \gamma)^2]\}^{1/2}. \end{aligned}$$

Finally, it will be convenient to denote the influence function of our estimator  $\hat{\lambda}$  by

$$\psi(Y, T, Z, X) \equiv b(Z, X)' \gamma (\nu_j(Y, T, Z, X) - b(Z, X)' \beta) + E_{\mu_{Z|X}}[b(Z, X)' \beta] - \lambda_{Q_0}, \quad (37)$$

and to let  $\sigma^2 \equiv \text{Var}_P\{\psi(Y, T, Z, X)\}$ . While we have suppressed it from the notation, we note that  $p$  (the dimension of  $b(Z, X)$ ) and  $j$  (as indexing  $\kappa_j$ ) can depend on  $n$ .

Given the introduced notation, we impose the following assumptions:

**ASSUMPTION 5.1:** (i)  $\{Y_i, T_i, X_i, Z_i\}_{i=1}^n$  is i.i.d.; (ii) There are known  $\{\nu_j\} \subseteq L^\infty(P)$  with  $\kappa_j \equiv \nu_j/\pi$  satisfying  $\Upsilon(\kappa_j) \rightarrow \ell$  in  $\tau$  topology; (iii)  $\mu_{Z|X} \ll P_{Z|X}$  and  $\|1/\pi\|_\infty < \infty$ .

**ASSUMPTION 5.2:** (i)  $\|b' \gamma\|_\infty = O(1)$  and  $B \equiv \|b' \beta\|_\infty \vee \|\nu_j\|_\infty < \infty$  satisfies  $B \log(n) = o(\sigma \sqrt{n})$ ; (ii)  $r_\beta \vee B r_\gamma \vee \sqrt{n} r_\beta r_\gamma = o_P(\sigma)$ ; (iii)  $\sqrt{n} \delta_\beta \delta_\gamma = o(\sigma)$ ; (iv)  $\sqrt{n} |\lambda_{Q_0} - E_P[\kappa_j(Y, T, Z, X)]| = o(\sigma)$ ; (v)  $|I_k| \asymp n$ .

By our identification analysis, Assumption 5.1(ii) is equivalent to identification of  $\lambda_{Q_0}$  in many applications. The condition that  $\nu_j$  be bounded can be relaxed by strengthening the norm under which  $\hat{\beta}_k$  and  $\hat{\gamma}_k$  converge. The requirements on  $\hat{\beta}_k$  and  $\hat{\gamma}_k$  are stated in Assumption 5.2. We impose high level conditions given the preponderance of results in

the literature justifying these assumptions under lower level assumptions; see, e.g., Chernozhukov et al. (2022b) and Chetverikov and Sørensen (2025). The rate requirement in Assumption 5.2(iii) is double robust in that it can hold even if  $\delta_\beta$  or  $\delta_\gamma$  not tend to zero. Assumption 5.2(iv) automatically holds when  $\{\kappa_j\}$  does not depend on  $j$  and may be viewed as an undersmoothing requirement otherwise.

Our next result establishes the asymptotic normality of our estimator.

**THEOREM 5.1:** *Let Assumptions 2.1, 2.2, 4.1, 5.1, 5.2 hold,  $\lambda_Q$  and  $\psi$  be as defined in (32) and (37), and  $\sigma^2 \equiv \text{Var}_P\{\psi(Y, T, Z, X)\}$ . Then, there is a  $\mathbb{Z} \sim N(0, 1)$  satisfying*

$$\frac{\sqrt{n}}{\sigma}(\hat{\lambda} - \lambda_{Q_0}) = \frac{1}{\sqrt{n}\sigma} \sum_{i=1}^n \psi(Y_i, T_i, Z_i, X_i) + o_P(1) = \mathbb{Z} + o_P(1). \quad (38)$$

For inference, we rely on a multiplier bootstrap procedure that readily extends to vector valued parameters and their functionals. Specifically, for each  $k$  we set

$$\hat{\psi}_k(Y, T, Z, X) \equiv b(Z, X)' \hat{\gamma}_k(\nu_j(Y, T, Z, X) - b(Z, X)' \hat{\beta}_k) + E_{\mu_{Z|X}}[b(Z, X)' \hat{\beta}_k] - \hat{\lambda}. \quad (39)$$

Our bootstrapped estimator  $\hat{\lambda}^*$  is then obtained by employing  $\hat{\psi}_k$  and an i.i.d. sample  $\{W_i\}_{i=1}^n$  of standard normal weights independent of the data to perturb  $\hat{\lambda}$  according to

$$\hat{\lambda}^* \equiv \hat{\lambda} + \frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} W_i \hat{\psi}_k(Y_i, T_i, Z_i, X_i).$$

We employ Gaussian weights for simplicity, though under appropriate moment restrictions the proposed bootstrap remains valid provided  $W$  satisfies  $E[W] = 0$  and  $E[W^2] = 1$ .

The next result establishes the validity of the proposed bootstrap.

**THEOREM 5.2:** *Let the conditions of Theorem 5.1 hold and  $\{W_i\}_{i=1}^n$  be i.i.d. standard normal random variables independent of  $\{Y_i, T_i, Z_i, X_i\}_{i=1}^n$ . Then, there exists a standard normal random variable  $\mathbb{Z}^*$  independent of  $\{Y_i, T_i, Z_i, X_i\}_{i=1}^n$  and satisfying*

$$\frac{\sqrt{n}}{\sigma}(\hat{\lambda}^* - \hat{\lambda}) = \frac{1}{\sqrt{n}\sigma} \sum_{i=1}^n W_i \psi(Y_i, T_i, Z_i, X_i) + o_P(1) = \mathbb{Z}^* + o_P(1). \quad (40)$$

Theorems 5.1, 5.2, and the Delta method justify employing the bootstrap to conduct inference on, e.g., conditional expectations of potential outcomes given types or of types

given covariates. These parameters equal  $F(\lambda_{1Q_0}, \dots, \lambda_{qQ_0})$  for some known differentiable function  $F$  and each  $\lambda_{iQ_0}$  with the structure in (32). For instance, to obtain a two-sided confidence region we: (i) Compute estimators  $(\hat{\lambda}_1, \dots, \hat{\lambda}_q)$  for  $(\lambda_{1Q_0}, \dots, \lambda_{qQ_0})$ ; (ii) Draw  $B$  samples  $\{W_i^{(b)}\}_{i=1}^n$  independent of the data; (iii) Employ each  $\{W_i^{(b)}\}_{i=1}^n$  to obtain bootstrap estimators  $(\hat{\lambda}_1^{*(b)}, \dots, \hat{\lambda}_q^{*(b)})$ ; (iv) Set  $\hat{c}_\alpha$  to equal the  $1 - \alpha$  quantile of  $\{|F(\hat{\lambda}_1^{*(b)}, \dots, \hat{\lambda}_q^{*(b)}) - F(\hat{\lambda}_1, \dots, \hat{\lambda}_q)|\}_{b=1}^B$ ; and (v) Report  $F(\hat{\lambda}_1, \dots, \hat{\lambda}_q) \pm \hat{c}_\alpha$  as a two-sided confidence region. Our results also allow us to conduct inference when  $F$  is directionally differentiable by using the framework in Fang and Santos (2018).

We conclude by illustrating our results in the context of Examples 2.4 and 2.5.

**Example 2.4 (cont.)** Recall that in this example  $Z$  had discrete support  $\mathbf{Z} = \{z_1, \dots, z_q\}$  and the support of  $T^*$  was restricted to  $\mathbf{R}^* \equiv \{t_1^*, \dots, t_r^*\}$ . We consider estimation of

$$E_{Q_0}[\rho(Y^*(t)) | T^* \in A] \quad (41)$$

for some  $A \subseteq \mathbf{R}^*$ . In (29) we showed  $E_{Q_0}[\rho(Y^*(t)) 1\{T^* \in A\}]$  is identified if and only if

$$\min_{\{s_j\}_{j=1}^q \subset \mathbf{R}} \sum_{i=1}^r (1\{t_i^* \in A\} - \sum_{j=1}^q s_j 1\{t_i^*(z_j) = t\})^2 = 0. \quad (42)$$

For any minimizer  $\{s_j\}$  to (42) we may then estimate  $E_{Q_0}[\rho(Y^*(t)) 1\{T^* \in A\}]$  by setting

$$\nu(Y, T, Z, X) = \rho(Y) 1\{T = t\} \left( \sum_{j=1}^q s_j 1\{Z = z_j\} \right) \quad E_{\mu_{Z|X}}[b(Z, X_i)] = \sum_{j=1}^q b(z_j, X_i)$$

in (35) and (36). Letting  $\rho(Y) = 1$ , we obtain an estimator for  $E_{Q_0}[1\{T^* \in A\}]$ . We estimate (41) by dividing the estimator for  $E_{Q_0}[\rho(Y^*(t)) 1\{T^* \in A\}]$  by the estimator for  $E_{Q_0}[1\{T^* \in A\}]$ . In this example, Assumptions 5.1(ii)(iii) holds provided  $\rho$  is bounded and  $P(Z = z|X)$  is bounded away from zero for all  $z \in \mathbf{Z}$ , while Assumption 5.2(iv) is automatic because  $\nu$  does not depend on  $j$  and therefore neither does  $\kappa = \nu/\pi$ . ■

**Example 2.5 (cont.)** Recall  $T^*$  is represented by  $(K_0^*, K_1^*)$  (see (24)). We illustrate our estimator in the context of estimating the expectation of a function  $\ell$  of  $(K_0^*, X)$ . First suppose  $\ell(K_0^*, X)$  is differentiable in  $K_0^*$  on  $(\underline{w}, \bar{w})$  with derivative  $\ell'(K_0^*, X)$  and  $\ell(K_0^*, X) = 0$  when  $K_0^* \in \{\underline{w}, \bar{w}, +\infty\}$ . Result (25) implies  $E_{Q_0}[\ell(K_0^*, X)]$  can be estimated by setting

$$\nu(Y, T, C, W, X) = 1\{T = 1, C = 0\} \ell'(W, X) \quad (43)$$

$$E_{\mu_{Z|X}}[b(C, W, X_i)] = \sum_{c=0}^1 \int_{\underline{w}}^{\bar{w}} b(c, w, X_i) dw \quad (44)$$

in (35) and (36). Expectations of other functions of  $(K_0^*, X)$  can be estimated by approximating them with differentiable functions. For example, the expectation of  $\ell(K_0^*) = 1\{a \leq K_0^* \leq b\}$  with  $\underline{w} < a < b < \bar{w}$  can be approximated by the expectation of  $\ell_j(K_0^*) = F((b - K_0^*)/h_j) - F((a - K_0^*)/h_j)$  for some  $h_j \downarrow 0$  and  $F$  the c.d.f. of a compactly supported mean zero continuous random variable. In this case, (43) becomes

$$\nu_j(Y, T, C, W, X) = 1\{T = 1, C = 0\} \times \frac{1}{h_j} (F'(\frac{a - W}{h_j}) - F'(\frac{b - W}{h_j})). \quad (45)$$

Assumptions 5.1(ii)(iii) require  $h_j = o(1)$  and  $f_{W|C}(W|c, X)P(C = c|X)$  be bounded away from zero. Under regularity conditions,  $|\lambda_{Q_0} - E_P[\kappa_j(T, Z, X)]| = O(h_j^2)$  and  $B \asymp \sigma^2 \asymp 1/h_j$  so Assumptions 5.2(i)(iv) hold when  $\log^2(n)/(nh_j) = o(1)$  and  $nh_j^5 = o(1)$ . ■

## APPENDIX A: PROOFS FOR SECTION 4

Throughout, we let  $Q_V$  denote the marginal distribution of a random variable  $V$  under  $Q$  and  $Q_{V|W}$  the conditional distribution of  $V$  given  $W$ . Distributions  $Q$  for  $(Y^*, T^*, Z, X)$  are defined on a product  $\sigma$ -field generated by  $\mathcal{F}_{Y^*} \times \mathcal{F}_{T^*} \times \mathcal{F}_Z \times \mathcal{F}_X$ , where  $\mathcal{F}_V$  denotes the  $\sigma$ -field on which  $Q_V$  is defined. In the main text we implicitly assumed  $P_{Z|X}$ ,  $\bar{Q}_{Y^*|X}$ , and  $\bar{Q}_{Y^*(t)|X}$  are well defined. Since  $Y^*(t) \in \mathbf{R}$ , these conditional distributions exist provided  $\mathbf{X}$  and  $\mathbf{Z}$  are Souslin spaces (e.g., Euclidean spaces) and  $\mathcal{F}_Z$  and  $\mathcal{F}_X$  are Borel  $\sigma$ -algebras; see Corollary 10.4.6 in Bogachev (2007). For any  $(V_1, V_2, W)$  with distribution  $\lambda$ , we understand  $V_1 \perp\!\!\!\perp V_2|W$  under  $\lambda$  to mean that, for  $i \neq j \in \{1, 2\}$  and any bounded  $f$

$$E_\lambda[f(V_j, W)|V_i, W] = E_\lambda[f(V_j, W)|W]. \quad (46)$$

We note that if (46) holds for  $(i, j) = (1, 2)$ , then it holds for  $(j, i) = (2, 1)$ .

*Proof of Lemma 4.1.* Lemma 13.14 and Corollary 3.5 in Aliprantis and Border (2006) imply  $D_0 \equiv \{dQ/d\mu : Q \in \Theta_0\} \subset L^1(\mu)$  is separable under  $\|\cdot\|_{\mu,1}$ . Therefore, there is a countable

$\mathcal{D} \equiv \{Q_i\}_{i=1}^\infty \subseteq \Theta_0$  such that  $\{dQ_i/d\mu\}_{i=1}^\infty$  is dense in  $D_0$ . Next, for any  $2 \leq n < \infty$  let

$$\lambda_{in} = \begin{cases} 1 - \sum_{i=2}^n \lambda_{in} & \text{if } i = 1 \\ 2^{-i} / \max\{1, \|dQ_i/d\mu\|_{\mathbf{Q}}\} & \text{if } 2 \leq i \leq n \\ 0 & \text{if } i > n \end{cases} \quad (47)$$

noting that  $\sum_{i=1}^n \lambda_{in} = 1$  and  $\lambda_{in} > 0$  for any  $1 \leq i \leq n$  due to  $\sum_{i=1}^\infty 2^{-i} = 1$ . Moreover,  $f_n \equiv \sum_{i=1}^n \lambda_{in}(dQ_i/d\mu)$  belongs to  $\mathcal{Q}$  for all  $n$  by convexity of  $\mathcal{Q}$  and for any  $n < m$

$$\|f_n - f_m\|_{\mathbf{Q}} \leq |\lambda_{1n} - \lambda_{1m}| \left\| \frac{dQ_1}{d\mu} \right\|_{\mathbf{Q}} + \sum_{i=n+1}^\infty \frac{1}{2^i}. \quad (48)$$

Since  $1/2 < \lambda_{1n}$  and  $\lambda_{1n}$  is decreasing in  $n$ , the sequence  $\{\lambda_{1n}\}_{n=1}^\infty$  has a limit in  $\mathbf{R}$ . Hence, by (48),  $\{f_n\}_{n=1}^\infty$  is Cauchy in  $\mathbf{Q}$  and, since  $\mathbf{Q}$  is complete and  $\mathcal{Q} \subseteq \mathbf{Q}$  is closed,  $\|f_n - q_0\|_{\mathbf{Q}} = o(1)$  for some  $q_0 \in \mathcal{Q} \subseteq L^1(\mu)$ . Finally, we define  $\bar{Q}$  by setting  $\bar{Q}(A) \equiv \int_A q_0 d\mu$ . Since  $\|\cdot\|_{\mu,1} \lesssim \|\cdot\|_{\mathbf{Q}}$ , we have  $\|f_n - q_0\|_{\mu,1} = o(1)$  and therefore

$$\lim_{n \rightarrow \infty} \left| \int g d\bar{Q} - \sum_{i=1}^n \lambda_{in} \int g dQ_i \right| \leq \lim_{n \rightarrow \infty} \|g\|_\infty \int |q_0 - \sum_{i=1}^n \lambda_{in} \frac{dQ_i}{d\mu}| d\mu = 0 \quad (49)$$

for any bounded  $g$ . In particular, result (49) implies that  $\int d\bar{Q} = 1$  and  $0 \leq \bar{Q}(A) \leq 1$ .

We next show  $\bar{Q} \in \Theta_0$ . Note (49) and  $Q_i \in \Theta_0$  imply, for any  $t \in \mathbf{T}$  and (measurable)  $V$ ,

$$\begin{aligned} \bar{Q}(T^*(Z) = t, (Y^*(t), Z, X) \in V) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \lambda_{in} Q_i(T^*(Z) = t, (Y^*(t), Z, X) \in V) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \lambda_{in} P(T = t, (Y, Z, X) \in V) = P(T = t, (Y, Z, X) \in V), \end{aligned} \quad (50)$$

since  $\sum_{i=1}^n \lambda_{in} = 1$ , which implies  $\bar{Q}$  induces  $P$ . Next note that, for any bounded  $f$  and  $g$ ,

$$\begin{aligned} E_{\bar{Q}}[g(Z, X)f(Y^*, T^*, X)] &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \lambda_{in} E_{Q_i}[g(Z, X)E_{Q_i}[f(Y^*, T^*, X)|X]] \\ &= E_{\bar{Q}}[g(Z, X)(\lim_{n \rightarrow \infty} \sum_{i=1}^n \lambda_{in} E_{Q_i}[f(Y^*, T^*, X)|X])], \end{aligned} \quad (51)$$

where the first equality follows from (49) and  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $Q_i$ , and the second from  $Q_i \in \Theta_0$ ,  $\bar{Q}$  inducing  $P$  by (50), and the dominated convergence theorem. Since (51) holds for any bounded  $g$ , Definition 10.1.1 in Bogachev (2007) yields

$$E_{\bar{Q}}[f(Y^*, T^*, X)|X, Z] = \lim_{n \rightarrow \infty} \sum_{i=1}^n \lambda_{in} E_{Q_i}[f(Y^*, T^*, X)|X] = E_{\bar{Q}}[f(Y^*, T^*, X)|X], \quad (52)$$

where the second equality follows by applying the equalities in (51) evaluated at functions  $g$  of  $X$  only. Therefore,  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $\bar{Q}$  and, by the preceding results, we conclude  $\bar{Q} \in \Theta_0$ . Next, fix an arbitrary  $Q \in \Theta_0$  and a set  $A$  with  $Q(A) > 0$ , and note  $\|dQ/d\mu - dQ_k/d\mu\|_{\mu,1} < Q(A)/2$  for some  $k$  because  $\{dQ_i/d\mu\}_{i=1}^\infty$  is dense in  $D_0$ . Hence,

$$Q_k(A) \geq Q(A) - \left| \int_A \left( \frac{dQ}{d\mu} - \frac{dQ_k}{d\mu} \right) d\mu \right| \geq Q(A) - \left\| \frac{dQ}{d\mu} - \frac{dQ_k}{d\mu} \right\|_{\mu,1} > \frac{Q(A)}{2} \quad (53)$$

and therefore  $Q_k(A) > 0$ . Since  $\{\lambda_{kn}\}_{n=1}^\infty$  is bounded away from zero for  $n$  sufficiently large, results (49) and (53) imply  $\bar{Q}(A) > 0$ . Since  $A$  was arbitrary, it follows that  $Q \ll \bar{Q}$ .

To establish  $\Theta_0$  is convex, let  $Q_1, Q_2 \in \Theta_0$ ,  $\gamma \in [0, 1]$ , and define  $Q_\gamma \equiv \gamma Q_1 + (1 - \gamma) Q_2$ . Then note: (i)  $dQ_\gamma/d\mu = \gamma dQ_1/d\mu + (1 - \gamma) dQ_2/d\mu \in \mathcal{Q}$  by Assumption 2.2(iii); (ii)  $Q_\gamma$  induces  $P$  by the arguments in (50) applied with  $Q_\gamma$  in place of  $\bar{Q}$ ; and (iii)  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $Q_\gamma$  by the arguments in (51) and (52) applied with  $Q_\gamma$  in place of  $\bar{Q}$ . ■

*Proof of Lemma 4.2.* First note  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$  and Lemma 4.1 imply  $Q(\Upsilon(\kappa) = \ell) = 1$  for all  $Q \in \Theta_0$ . Therefore,  $\ell(Y^*, T^*, X) = E_Q[\kappa(Y, T, Z, X)|Y^*, T^*, X]$  for all  $Q \in \Theta_0$  by Corollary A.1, which implies  $\ell \in L^1(Q)$  by Jensen's inequality and  $\kappa \in L^1(P)$ . Moreover, for any  $Q \in \Theta_0$  we obtain  $E_Q[\ell(Y^*, T^*, X)] = E_Q[\kappa(Y, T, Z, X)] = E_P[\kappa(Y, T, Z, X)]$  by the law of iterated expectations, and hence the lemma follows. ■

*Proof of Corollary 4.1.* By Lemma A.2, we have  $\mu(\Upsilon(\kappa) = \ell) = 1$  and hence also that  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$  due to  $\bar{Q} \ll \mu$ . Thus, the result is immediate from Lemma 4.2. ■

*Proof of Theorem 4.1.* We first show  $\lambda_{Q_0}$  being identified implies  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$ . Let  $\mathcal{B}'$  denote the linear span of  $\mathcal{R} \cup \{\ell\}$  and note every  $B \in \mathcal{B}$  is a linear functional on  $\mathcal{B}'$ . We endow  $\mathcal{B}'$  with the weak topology generated by  $\mathcal{B}$ , denoted  $\sigma(\mathcal{B}', \mathcal{B})$ , and note  $(\mathcal{B}', \sigma(\mathcal{B}', \mathcal{B}))$  is a topological vector space whose dual is  $\mathcal{B}$ . Moreover, Theorem 5.73 in Aliprantis and Border (2006) implies  $(\mathcal{B}', \sigma(\mathcal{B}', \mathcal{B}))$  is locally convex, while Lemma 2.53 in Aliprantis and Border (2006) implies the  $\tau$  topology on  $\mathcal{R} \cup \{\ell\}$  equals the relative

topology on  $\mathcal{R} \cup \{\ell\}$  induced by the topology  $\sigma(\mathcal{B}', \mathcal{B})$  on  $\mathcal{B}'$ . Therefore,  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$  (in  $\mathcal{R} \cup \{\ell\}$ ) if and only if  $\ell$  belongs to the  $\sigma(\mathcal{B}', \mathcal{B})$ -closure of  $\mathcal{R}$  in  $\mathcal{B}'$ , denoted  $\bar{\mathcal{R}}$ . We show  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$  by showing  $\ell \in \bar{\mathcal{R}}$ .

We proceed by contradiction and suppose  $\ell \notin \bar{\mathcal{R}}$ . Since  $\mathcal{B}$  is the dual of  $(\mathcal{B}', \sigma(\mathcal{B}', \mathcal{B}))$ , Corollary 5.80 in Aliprantis and Border (2006) implies there is  $B_0 \in \mathcal{B}$  satisfying  $B_0(\ell) \neq 0$  and  $B_0(B') = 0$  for all  $B' \in \bar{\mathcal{R}}$ . Moreover, by definition of  $\mathcal{B}$  there is  $\{(s_j, Q_j)\}_{j=1}^J$  with  $s_j \in \mathcal{S}_{Q_j}$  and  $Q_j \in \Theta_0$  for all  $1 \leq j \leq J < \infty$ , and  $B_0(B') = B'(\sum_{j=1}^J \langle \cdot, s_j \rangle_{Q_j})$  for all  $B' \in \mathcal{B}'$ . Hence,  $B_0(B') = 0$  for all  $B' \in \bar{\mathcal{R}}$  implies, for any  $f \in L^1(P)$ , that

$$\begin{aligned} 0 &= \sum_{j=1}^J E_{Q_j} \left[ \left( \sum_{t \in \mathbf{T}} E_{P_{Z|X}} [f(Y^*(t), t, Z, X) 1\{T^*(Z) = t\}] \right) s_j(Y^*, T^*, X) \right] \\ &= E_P [f(Y, T, Z, X) \left( \sum_{j=1}^J E_{Q_j} [s_j(Y^*, T^*, X) | Y, T, Z, X] \right)], \end{aligned} \quad (54)$$

where the second equality follows from Corollary A.1, the law of iterated expectations, and  $Q_j \in \Theta_0$ . It follows  $P(\sum_{j=1}^J E_{Q_j} [s_j(Y^*, T^*, X) | Y, T, Z, X] = 0) = 1$  because (54) holds for all  $f \in L^1(P)$ . Therefore, by Lemma A.3 there are  $\tilde{Q}, Q^a \in \Theta_0$  and  $\eta > 0$  such that  $\tilde{Q}(A) = Q^a(A) + \eta \sum_{j=1}^J E_{Q_j} [s_j(Y^*, T^*, X) 1\{(Y^*, T^*, Z, X) \in A\}]$  for all  $A$ . Letting  $\mathbf{1}$  denote the constant function taking value one, we then obtain

$$\lambda_{\tilde{Q}} = \langle \ell, \mathbf{1} \rangle_{Q^a} + \eta \sum_{j=1}^J \langle \ell, s_j \rangle_{Q_j} = \lambda_{Q^a} + \eta L_0(\ell).$$

However,  $\eta > 0$  and  $B_0(\ell) \neq 0$  together imply that  $\lambda_{\tilde{Q}} \neq \lambda_{Q^a}$ . Thus, since  $\tilde{Q}, Q^a \in \Theta_0$  we obtain that  $\lambda_{Q_0}$  is not identified, reaching a contradiction.

For the converse, suppose  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$ . Since  $\mathcal{B}$  is identified (because it only depends on  $\Theta_0$ ),  $\ell$  is identified, and  $\mathcal{R}$  is identified (because  $\Upsilon : L^1(P) \rightarrow L^1(\bar{Q})$  is identified), Theorem 2.4 in Aliprantis and Border (2006) implies there is an identified net  $\{f_\alpha\}_{\alpha \in \mathcal{A}} \subset \mathcal{R}$  satisfying  $\lim_\alpha B(f_\alpha) = B(\ell)$  for all  $B \in \mathcal{B}$ . Hence, for any  $Q_1, Q_2 \in \Theta_0$

$$\lambda_{Q_1} = \langle \ell, \mathbf{1} \rangle_{Q_1} = \lim_\alpha \langle f_\alpha, \mathbf{1} \rangle_{Q_1} = \lim_\alpha \langle f_\alpha, \mathbf{1} \rangle_{Q_2} = \langle \ell, \mathbf{1} \rangle_{Q_2} = \lambda_{Q_2},$$

where the first and last equalities follow by definition of  $\lambda_Q$ , the second and fourth equalities from  $\lim_{\alpha} B(f_{\alpha}) = B(\ell)$  for all  $B \in \mathcal{B}$ , and the third equality by Lemma 4.2 and  $f_{\alpha} \in \mathcal{R}$ . Thus, we conclude that  $\lambda_Q$  is constant in  $Q \in \Theta_0$  and that  $\lambda_{Q_0}$  is identified. ■

*Proof of Corollary 4.2.* To apply Theorem 4.1 note Assumptions 2.1 and 2.2 are imposed. Moreover, Assumption 4.1(i) holds due to  $\ell \in L^1(\mu)$ ,  $dQ/d\mu \in L^{\infty}(\mu)$  for all  $Q \in \Theta_0$  by Assumption 2.2(iii), and Holder's inequality implying  $\int |\ell| dQ \leq \|\ell\|_{\mu,1} \|\frac{dQ}{d\mu}\|_{\mu,\infty} < \infty$  for any  $Q \in \Theta_0$ . Assumption 4.1(ii) is immediate because  $\mathcal{Q} = \mathbf{Q}$ . Thus, Theorem 4.1 implies that  $\lambda_Q$  is identified if and only if  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$ .

To establish part (i), note  $\ell \in L^1(Q)$  for all  $Q \in \Theta_0$  and  $sd\bar{Q}/d\mu \in L^{\infty}(\mu)$  for any  $s \in L^{\infty}(\bar{Q}_{Y^*T^*X})$  (because  $\bar{Q} \ll \mu$  and  $d\bar{Q}/d\mu \in L^{\infty}(\mu)$ ) imply  $\mathcal{S}_{\bar{Q}} = L^{\infty}(\bar{Q}_{Y^*T^*X})$ . Therefore, if  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$ , then  $\|\ell - \Upsilon(\kappa_j)\|_{\bar{Q},1} = o(1)$  for some  $\{\kappa_j\} \subseteq L^1(P)$  by Lemma A.4. Since  $\mu \ll \bar{Q}$  and  $d\mu/d\bar{Q}$  is bounded,  $\{\kappa_j\}$  satisfies  $\|\ell - \Upsilon(\kappa_j)\|_{\mu,1} = o(1)$ . For the converse, note that if  $\|\ell - \Upsilon(\kappa_j)\|_{\mu,1} = o(1)$ , then  $\|\ell - \Upsilon(\kappa_j)\|_{Q,1} = o(1)$  for all  $Q \in \Theta_0$  due to Holder's inequality and  $dQ/d\mu \in L^{\infty}(\mu)$ . Hence, for any  $Q \in \Theta_0$ ,  $\int \ell dQ = \int \Upsilon(\kappa_j) dQ + o(1)$  and Lemma 4.2 implies  $\lambda_Q = E_P[\kappa_j(Y, T, Z, X)] + o(1)$ . Part (ii) follows from part (i) and Lemma A.2. ■

*Proof of Theorem 4.2.* By Theorem 4.1,  $\lambda_{Q_0}$  is identified if and only if  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$ . Since  $\mathcal{R}_T \subseteq \mathcal{R}$ ,  $\lambda_{Q_0}$  is identified when  $\ell$  is in the  $\tau$ -closure of  $\mathcal{R}_T$ . We show the converse by showing that if  $\ell$  is in the  $\tau$ -closure of  $\mathcal{R}$ , then it belongs to the  $\tau$ -closure of  $\mathcal{R}_T$ . If  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}$ , then Theorem 2.14 in Aliprantis and Border (2006) implies that there is a net  $\{f_{\alpha}\}_{\alpha \in \mathcal{A}} \subseteq L^1(P)$  satisfying

$$\lim_{\alpha} \langle s, \Upsilon(f_{\alpha}) \rangle_Q = \langle s, \ell \rangle_Q \quad (55)$$

for all  $s \in \mathcal{S}_Q$ ,  $Q \in \Theta_0$ . Set  $g_{\alpha}(t, Z, X) \equiv E_{\bar{Q}_{Y^*|X}}[f_{\alpha}(Y^*(t), t, Z, X)]$  for any  $t \in \mathbf{T}$ . By Jensen's inequality,  $\bar{Q} \in \Theta_0$ , Lemma A.1,  $\bar{Q}_{ZX}^{\text{it}} = \bar{Q}_{ZX}$ , and  $\bar{Q}_{T^*X}^{\text{it}} = \bar{Q}_{T^*X}$  we have

$$\begin{aligned} E_P[|g_{\alpha}(T, Z, X)|] &= E_{\bar{Q}_{T^*ZX}} \left[ \sum_{t \in \mathbf{T}} 1\{T^*(Z) = t\} |E_{\bar{Q}_{Y^*|X}}[f_{\alpha}(Y^*(t), t, Z, X)]| \right] \\ &\leq E_{\bar{Q}_{Y^*T^*X}^{\text{it}}} \left[ \sum_{t \in \mathbf{T}} E_{\bar{Q}_{Z|X}}[1\{T^*(Z) = t\} |f_{\alpha}(Y^*(t), t, Z, X)|] \right]. \end{aligned} \quad (56)$$

Since the constant function taking value one belongs to  $\mathcal{S}_{\bar{Q}}$ , Assumption 4.2(ii) implies that  $d\bar{Q}_{Y^*T^*X}^{\text{it}}/d\bar{Q}_{Y^*T^*X} \in \mathcal{S}_{\bar{Q}} \subseteq L^{\infty}(\bar{Q}_{Y^*T^*X})$ . Therefore,  $d\bar{Q}_{Y^*T^*X}^{\text{it}}/d\bar{Q}_{Y^*T^*X}$  is

bounded, which together with (56), Corollary A.1, and  $\bar{Q} \in \Theta_0$  imply  $E_P[|g_\alpha(T, Z, X)|] \lesssim E_P[|f_\alpha(Y, T, Z, X)|]$ . Thus, since  $f_\alpha \in L^1(P)$ , it follows  $g_\alpha \in L^1(P_{TZ X})$ .

For any  $s_0 \in \mathcal{S}_{\bar{Q}} \cap L^\infty(\bar{Q}_{T^* X})$  note  $\bar{Q}_{Z|X}^{\text{it}} = \bar{Q}_{Z|X} = P_{Z|X}$  (because  $\bar{Q} \in \Theta_0$ ) implies

$$\begin{aligned} \langle s_0, \Upsilon(g_\alpha) \rangle_{\bar{Q}} &= E_{\bar{Q}_{T^* X}}[E_{P_{Z|X}}[E_{\bar{Q}_{Y^*|X}}[\sum_{t \in \mathbf{T}} 1\{T^*(Z) = t\} f_\alpha(Y^*(t), t, Z, X) s_0(T^*, X)]]] \\ &= E_{\bar{Q}_{Y^* T^* X}^{\text{it}}}[\sum_{t \in \mathbf{T}} E_{\bar{Q}_{Z|X}^{\text{it}}}[1\{T^*(Z) = t\} f_\alpha(Y^*(t), t, Z, X) s_0(T^*, X)]]]. \end{aligned} \quad (57)$$

Moreover, since  $s_0$  and  $\ell$  only depend on  $(T^*, X)$ , it follows from  $\bar{Q}_{T^* X} = \bar{Q}_{T^* X}^{\text{it}}$  that

$$\begin{aligned} \langle \ell, s_0 \rangle_{\bar{Q}} &= \langle \ell, s_0 \rangle_{\bar{Q}_{T^* X}} = \langle \ell, s_0 \rangle_{\bar{Q}_{T^* X}^{\text{it}}} = \langle \ell, s_0 \frac{d\bar{Q}_{Y^* T^* X}^{\text{it}}}{d\bar{Q}_{Y^* T^* X}} \rangle_{\bar{Q}} \\ &= \lim_{\alpha} \langle \Upsilon(f_\alpha), s_0 \frac{d\bar{Q}_{Y^* T^* X}^{\text{it}}}{d\bar{Q}_{Y^* T^* X}} \rangle_{\bar{Q}} = \lim_{\alpha} \langle \Upsilon(f_\alpha), s_0 \rangle_{\bar{Q}^{\text{it}}} = \lim_{\alpha} \langle \Upsilon(g_\alpha), s_0 \rangle_{\bar{Q}}, \end{aligned} \quad (58)$$

where the fourth equality follows from (55),  $\bar{Q} \in \Theta_0$ , and  $s_0(d\bar{Q}_{Y^* T^* X}^{\text{it}}/d\bar{Q}_{Y^* T^* X}) \in \mathcal{S}_{\bar{Q}}$  by Assumption 4.2(ii), and the sixth from result (57) and  $\bar{Q}_{Z|X}^{\text{it}} = \bar{Q}_{Z|X} = P_{Z|X}$ . Next set  $\Pi_Q(s)(T^*, X) \equiv E_Q[s(Y^*, T^*, X)|T^*, X]$  and note for any  $s \in \mathcal{S}_Q$  and  $Q \in \Theta_0$  we have

$$\langle \ell, s \rangle_Q = \langle \ell, \Pi_Q(s) \frac{dQ_{T^* X}}{d\bar{Q}_{T^* X}} \rangle_{\bar{Q}} = \lim_{\alpha} \langle \Upsilon(g_\alpha), \Pi_Q(s) \frac{dQ_{T^* X}}{d\bar{Q}_{T^* X}} \rangle_{\bar{Q}} = \lim_{\alpha} \langle \Upsilon(g_\alpha), s \rangle_Q, \quad (59)$$

where the first equality follows from  $\ell$  depending only on  $(T^*, X)$ , and the second from (58) and  $\Pi_Q(s)dQ_{T^* X}/d\bar{Q}_{T^* X} \in \mathcal{S}_{\bar{Q}}$  by Assumptions 4.2(iii)(iv). Result (59) and Theorem 2.14 in Aliprantis and Border (2006) then imply  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}_T$ . ■

*Proof of Corollary 4.3.* If  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$  for some  $\kappa \in L^1(P_{TZ X})$ , then  $\lambda_{Q_0}$  is identified by Lemma 4.2. To establish the converse, we verify the conditions for Theorem 4.2. Note Assumptions 2.1 and 2.2 are assumed, and Assumption 4.1(ii) holds because  $\mathcal{Q} = \mathbf{Q} = L^1(\mu)$ . Define  $\mathbf{T}_0^* \equiv \{t^* \in \mathbf{T}^* : \bar{Q}(T^* = t^*) > 0\}$  and note that  $Q(T^* \in \mathbf{T}_0^*) = 1$  for any  $Q \in \Theta_0$  due to  $Q \ll \bar{Q}$ . Since  $Q_X = \bar{Q}_X = P_X$  for any  $Q \in \Theta_0$ , it follows that

$$\frac{dQ_{T^* X}}{d\bar{Q}_{T^* X}}(T^*, X) = \sum_{t^* \in \mathbf{T}_0^*} 1\{T^* = t^*\} \frac{Q(T^* = t^*|X)}{\bar{Q}(T^* = t^*|X)} \leq \frac{1}{\varepsilon} \quad (60)$$

because  $\bar{Q}(T^* = t^*|X) \geq \varepsilon$  almost surely. Hence,  $\ell \in L^1(\bar{Q}_{T^* X})$  and (60) imply  $\ell \in L^1(Q)$  for any  $Q \in \Theta_0$ , verifying Assumption 4.1(i). Moreover, since  $\mathcal{S}_Q = L^\infty(Q_{Y^* T^* X})$  for any

$Q \in \Theta_0$  and  $d\bar{Q}^{\text{it}}/d\bar{Q}$  is bounded, it follows from (60) that Assumption 4.2 also holds.

Thus, we may apply Theorem 4.2 together with Lemma A.4 to conclude that if  $\lambda_{Q_0}$  is identified, then there exists  $\{\kappa_j\}_{j=1}^\infty \subseteq L^1(P_{TZ X})$  satisfying  $\|\ell - \Upsilon(\kappa_j)\|_{\bar{Q},1} = o(1)$ .

Next let  $\mathbf{S}_0 \equiv \{(t, z) \in \mathbf{T} \times \mathbf{Z} : P(T = t, Z = z) > 0\}$  and note that for any  $(t, z) \in \mathbf{S}_0$

$$P(T = t, Z = z | X) = \sum_{t^* \in \mathbf{T}_0^* : t^*(z) = t} \bar{Q}(T^* = t^* | X) P(Z = z | X) \geq \varepsilon^2 \quad (61)$$

because  $\bar{Q} \in \Theta_0$ ,  $(Y^*, T^*) \perp\!\!\!\perp Z | X$ ,  $\bar{Q}(T^* = t^* | X) \geq \varepsilon$  for any  $t^* \in \mathbf{T}_0^*$ , and  $P(Z = z | X) \geq \varepsilon$  for any  $z \in \mathbf{Z}$ . Hence, for any event  $E$  with  $P(X \in E) > 0$  and  $(t, z) \in \mathbf{S}_0$ ,

$$P(X \in E) = \frac{P(X \in E | T = t, Z = z) P(T = t, Z = z)}{P(T = t, Z = z | X \in E)} \leq \frac{P(X \in E | T = t, Z = z)}{\varepsilon^2}. \quad (62)$$

Let  $P_{X|t,z}$  denote the distribution of  $X$  conditional on  $(T, Z) = (t, z)$  for any  $(t, z) \in \mathbf{S}_0$ .

Then note (62) implies  $dP_X/dP_{X|t,z} \leq \varepsilon^{-2}$ , which together with (61) yields  $\kappa_j(t, z, \cdot) \in L^1(P_X)$  for any  $(t, z) \in \mathbf{S}_0$ . Next set  $\tilde{\kappa}_j(t, z, X) \equiv \kappa_j(t, z, X) P(Z = z | X)$  and note

$$\|\ell - \Upsilon(\kappa_j)\|_{\bar{Q},1} = E_{\bar{Q}}[\|\ell(T^*, X) - \sum_{(t,z) \in \mathbf{S}_0} 1\{T^*(z) = t\} \tilde{\kappa}_j(t, z, X)\|]. \quad (63)$$

Since  $\tilde{\kappa}_j(t, z, \cdot) \in L^1(P_X)$  for all  $(t, z) \in \mathbf{S}_0$ ,  $\|\ell - \Upsilon(\kappa_j)\|_{\bar{Q},1} = o(1)$ , (63), and Lemma A.5 imply there are functions  $\{f_0(t, z, X)\}_{(t,z) \in \mathbf{S}_0}$  satisfying  $f_0(t, z, \cdot) \in L^1(P_X)$  and

$$E_{\bar{Q}}[\|\ell(T^*, X) - \sum_{(t,z) \in \mathbf{S}_0} 1\{T^*(z) = t\} f_0(t, z, X)\|] = 0. \quad (64)$$

Finally, set  $\kappa(t, z, X) \equiv f_0(t, z, X)/P(Z = z | X)$  and note  $\kappa \in L^1(P_{TZ X})$  because  $P(Z = z | X) \geq \varepsilon > 0$  and  $f_0(t, z, \cdot) \in L^1(P_X)$ . By result (63) and direct calculation,  $\|\ell - \Upsilon(\kappa)\|_{\bar{Q},1} = 0$ , yielding that identification of  $\lambda_Q$  implies the existence of the desired  $\kappa$  and hence establishing part (i) of the corollary. Part (ii) is immediate from part (i), Lemma A.2,  $\mu \ll \bar{Q}$  by assumption, and  $\bar{Q} \ll \mu$  since  $\bar{Q} \in \Theta_0$ . ■

*Proof of Theorem 4.3.* We first show that if  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}_t$ , then  $\lambda_{Q_0}$  is identified. To this end, note  $\mathcal{B}$  and  $\mathcal{R}_t$  are identified because  $\Theta_0$  and  $\Upsilon : L^1(P) \rightarrow L^1(\bar{Q})$  are identified. Hence, Theorem 2.14 in Aliprantis and Border (2006) and  $\ell$  being in the  $\tau$ -closure of  $\mathcal{R}_t$  imply there is an identified net  $\{f_\alpha\}_{\alpha \in \mathcal{A}} \subseteq L^1(P_{TZ X})$  satisfying

$$\lim_{\alpha} \langle \Upsilon(f_\alpha), s \rangle_Q = \langle \ell, s \rangle_Q \text{ for all } s \in \mathcal{S}_Q, Q \in \Theta_0. \quad (65)$$

Next, let  $\Pi_Q(s)(T^*, X) \equiv E_Q[s(Y^*, T^*, X)|T^*, X]$  for any  $s \in L^1(Q)$ , and note that  $\lambda_Q = \langle \ell, \Pi_Q(\rho) \rangle_Q$ . Therefore, for any  $Q \in \Theta_0$ ,  $\Pi_Q(\rho) \in \mathcal{S}_Q$  by Assumption 4.4(i), result (65), and  $f_\alpha(T, Z, X) = 1\{T = t\}g_\alpha(Z, X)$  for some  $g_\alpha$  by definition of  $\mathcal{R}_t$  imply

$$\begin{aligned} \lambda_Q &= \lim_{\alpha} E_Q[E_Q[\rho(Y^*(t), X)|T^*, X]E_{P_{Z|X}}[g_\alpha(Z, X)1\{T^*(Z) = t\}]] \\ &= \lim_{\alpha} E_P[\rho(Y, X)g_\alpha(Z, X)1\{T = t\}], \end{aligned} \quad (66)$$

where the final equality follows from Corollary A.1, the law of iterated expectations, and  $Q \in \Theta_0$ . Since (66) holds for any  $Q \in \Theta_0$ , it follows that  $\lambda_{Q_0}$  is identified.

We next show  $\lambda_{Q_0}$  being identified implies  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}_t$ . To this end, we first note that Theorem 4.1 and  $\lambda_{Q_0}$  being identified imply  $\ell\rho$  belongs to the  $\tau$ -closure of  $\mathcal{R}$ . By Theorem 2.14 in Aliprantis and Border (2006), then

$$\lim_{\alpha} \langle \Upsilon(v_\alpha), s \rangle_Q = \langle \ell\rho, s \rangle_Q \text{ for all } s \in \mathcal{S}_Q, Q \in \Theta_0 \quad (67)$$

for some net  $\{v_\alpha\}_{\alpha \in \mathcal{A}} \subseteq L^1(P)$ . Moreover,  $Y^*(t) \perp\!\!\!\perp T^*|X$  under  $\bar{Q}^{\text{io}}$ ,  $\bar{Q}_{Y^*(t)X} = \bar{Q}_{Y^*(t)X}^{\text{io}}$  and Assumptions 4.4(ii)(iii) imply  $\bar{Q}^{\text{io}}(E_{\bar{Q}^{\text{io}}}[\rho(Y^*(t), X)\phi_{\bar{Q},\rho}(Y^*(t), X)|T^*, X] = 1) = 1$ . Therefore, for any  $s \in \mathcal{S}_{\bar{Q}}$ , the law of iterated expectations and  $\bar{Q}_{Y^*(t)X} = \bar{Q}_{Y^*(t)X}^{\text{io}}$  imply

$$\langle \ell, s \rangle_{\bar{Q}} = \langle \ell\rho, \phi_{\bar{Q},\rho}\Pi_{\bar{Q}}(s) \rangle_{\bar{Q}^{\text{io}}} = \langle \ell\rho, \phi_{\bar{Q},\rho}\Pi_{\bar{Q}}(s) \frac{d\bar{Q}_{Y^*T^*X}^{\text{io}}}{d\bar{Q}_{Y^*T^*X}} \rangle_{\bar{Q}}, \quad (68)$$

where the final equality used Assumption 4.4(ii). Moreover, since 4.4(i)(iii) imply that  $\phi_{\bar{Q},\rho}\Pi_{\bar{Q}}(s)(d\bar{Q}_{Y^*T^*X}^{\text{io}}/d\bar{Q}_{Y^*T^*X}) \in \mathcal{S}_{\bar{Q}}$  for any  $s \in \mathcal{S}_{\bar{Q}}$ , we have

$$\langle \ell, s \rangle_{\bar{Q}} = \lim_{\alpha} \langle \Upsilon(v_\alpha), \phi_{\bar{Q},\rho}\Pi_{\bar{Q}}(s) \frac{d\bar{Q}_{Y^*T^*X}^{\text{io}}}{d\bar{Q}_{Y^*T^*X}} \rangle_{\bar{Q}} = \lim_{\alpha} \langle \Upsilon(v_\alpha), \phi_{\bar{Q},\rho}\Pi_{\bar{Q}}(s) \rangle_{\bar{Q}^{\text{io}}} \quad (69)$$

due to (67) and (68). Next set  $g_\alpha(Z, X) \equiv E_{\bar{Q}_{Y^*(t)|X}}[v_\alpha(Y^*(t), t, Z, X)\phi_{\bar{Q},\rho}(Y^*(t), X)]$  and  $f_\alpha(T, Z, X) \equiv 1\{T = t\}g_\alpha(Z, X)$ . Also note that  $\bar{Q} \in \Theta_0$ , Jensen's inequality,  $\phi_{\bar{Q},\rho} \in L^\infty(\bar{Q})$  by Assumption 4.4(iii), and the definition of  $\bar{Q}^{\text{io}}$  imply that

$$\begin{aligned} E_P[|f_\alpha(T, Z, X)|] &\lesssim E_{\bar{Q}_{T^*ZX}}[1\{T^*(Z) = t\}E_{\bar{Q}_{Y^*(t)|X}}[|v_\alpha(Y^*(t), t, Z, X)|]] \\ &= E_{\bar{Q}^{\text{io}}}[1\{T^*(Z) = t\}|v_\alpha(Y^*(t), t, Z, X)|] \lesssim E_P[1\{T = t\}v_\alpha(Y, T, Z, X)], \end{aligned} \quad (70)$$

where the final inequality follows from  $\bar{Q} \in \Theta_0$  and  $d\bar{Q}^{\text{io}}/d\bar{Q} \in L^\infty(\bar{Q})$  by Assumption 4.4(ii). Since  $v_\alpha \in L^1(P)$ , (70) implies  $\Upsilon(f_\alpha) \in \mathcal{R}_t$ . Also, the law of iterated expectations,  $\bar{Q} \in \Theta_0$ , Corollary A.1,  $\bar{Q}_{T^*ZX}^{\text{io}} = \bar{Q}_{T^*ZX}$ , and the definition of  $f_\alpha$  imply for any  $s \in \mathcal{S}_{\bar{Q}}$

$$\begin{aligned} & \langle \Upsilon(f_\alpha), s \rangle_{\bar{Q}} \\ &= E_{\bar{Q}^{\text{io}}} [v_\alpha(Y^*(t), t, Z, X) 1\{T^*(Z) = t\} \phi_{\bar{Q}, \rho}(Y^*(t), X) E_{\bar{Q}}[s(Y^*, T^*, X) | T^*, X]] \\ &= \sum_{\tilde{t} \in \mathbf{T}} E_{\bar{Q}^{\text{io}}} [v_\alpha(Y^*(\tilde{t}), \tilde{t}, Z, X) 1\{T^*(Z) = \tilde{t}\} \phi_{\bar{Q}, \rho}(Y^*(t), X) E_{\bar{Q}}[s(Y^*, T^*, X) | T^*, X]] \\ &= \langle \Upsilon(v_\alpha), \phi_{\bar{Q}, \rho} \Pi_{\bar{Q}}(s) \rangle_{\bar{Q}^{\text{io}}}, \end{aligned} \tag{71}$$

where the second equality follows from  $(Y^*(\tilde{t}), T^*, Z)$  being independent of  $Y^*(t)$  conditionally on  $X$  under  $\bar{Q}^{\text{io}}$  whenever  $\tilde{t} \neq t$ ,  $E_{\bar{Q}^{\text{io}}}[\phi_{\bar{Q}, \rho}(Y^*(t), X) | X] = 0$  and  $\bar{Q}_{Y^*(t)X}^{\text{io}} = \bar{Q}_{Y^*(t)X}$ ; and the final equality holds by Lemma A.1 and  $\bar{Q}_{ZX}^{\text{io}} = \bar{Q}_{ZX} = P_{ZX}$ . To conclude, note that for any  $Q \in \Theta_0$  and  $s \in \mathcal{S}_Q$ ,  $Q \ll \bar{Q}$ ,  $\Pi_Q(s) dQ_{T^*X}/d\bar{Q}_{T^*X} \in \mathcal{S}_{\bar{Q}}$  by Assumptions 4.4(i)(iv), and results (69) and (71) yield that

$$\langle \ell, s \rangle_Q = \langle \ell, \Pi_Q(s) \frac{dQ_{T^*X}}{d\bar{Q}_{T^*X}} \rangle_{\bar{Q}} = \lim_{\alpha} \langle \Upsilon(f_\alpha), \Pi_Q(s) \frac{dQ_{T^*X}}{d\bar{Q}_{T^*X}} \rangle_{\bar{Q}} = \lim_{\alpha} \langle \Upsilon(f_\alpha), s \rangle_Q.$$

Hence, since  $\Upsilon(f_\alpha) \in \mathcal{R}_t$ , we conclude that  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{R}_t$ . ■

*Proof of Corollary 4.4.* The proof is similar to that of Corollary 4.3 and we therefore omit details. In particular, the same arguments used in Corollary 4.4 here imply that the conditions of Theorem 4.3 are satisfied. Therefore, if (i) holds, then Theorem 4.3 and Lemma A.4 yield that there is  $\{\kappa_j\} \in L^1(P_{ZXt})$  satisfying  $\|\ell - \Upsilon(\kappa_j)\|_{\bar{Q},1} = o(1)$ .

Next, let  $\mathbf{Z}_0 \equiv \{z \in \mathbf{Z} : P(T = t, Z = z) > 0\}$ , note  $\kappa_j(T, Z, X) = 1\{T = t\} g_j(Z, X)$  for some  $g_j$ , and use the same arguments in Corollary 4.3 to show  $g_j(z, \cdot) \in L^1(P_X)$  for any  $z \in \mathbf{Z}_0$ . Setting  $\tilde{g}_j(z, X) \equiv g_j(z, X) P(Z = z | X)$ , we obtain by definition of  $\Upsilon$

$$\|\ell - \Upsilon(\kappa_j)\|_{\bar{Q},1} = E_{\bar{Q}}[|\ell(T^*, X) - \sum_{z \in \mathbf{Z}_0} 1\{T^*(z) = t\} \tilde{g}_j(z, X)|]. \tag{72}$$

Therefore,  $\|\ell - \Upsilon(\kappa_k)\|_{\bar{Q},1} = o(1)$ , result (72), and Lemma A.5 allow us to conclude

$$E_{\bar{Q}}[|\ell(T^*, X) - \sum_{z \in \mathbf{Z}_0} 1\{T^*(z) = t\} f_0(z, X)|] = 0 \tag{73}$$

for some  $\{f_0(z, X)\}_{z \in \mathbf{Z}_0}$  satisfying  $f_0(z, \cdot) \in L^1(P_X)$  for all  $z \in \mathbf{Z}_0$ . Hence, setting  $\kappa(T, Z, X) \equiv 1\{T = t\} \sum_{z \in \mathbf{Z}_0} 1\{Z = z\} f_0(Z, X) / P(Z = z|X)$  we obtain from  $P(Z = z|X) \geq \varepsilon > 0$  that  $\kappa \in L^1(P_{tZX})$  and from (73) that  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$ . Since  $\bar{Q}(\Upsilon(\kappa) = \ell) = 1$  implies  $\bar{Q}(\Upsilon(f\kappa) = f\ell) = 1$  for any bounded  $f$ , Lemma 4.2 yields that (i) implies (iii). Because (iii) implies (ii) and (ii) implies (i), the corollary follows. ■

**COROLLARY A.1:** *If Assumption 2.1 holds,  $Q \in \Theta_0$ , and  $f \in L^1(P)$ , then it follows*

$$E_Q[f(Y, T, Z, X)|Y^*, T^*, X] = \sum_{t \in \mathbf{T}} E_{P_{Z|X}}[f(Y^*(t), t, Z, X) 1\{T^*(Z) = t\}].$$

*Proof.* Follows from Lemma A.1,  $Q_{Z|X} = P_{Z|X}$  due to  $Q \in \Theta_0$ , and  $Y = Y^*(T)$  and  $T = T^*(Z)$  implying that  $f(Y, T, Z, X) = \sum_{t \in \mathbf{T}} f(Y^*(t), t, Z, X) 1\{T^*(Z) = t\}$ . ■

**LEMMA A.1:** *If  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $Q$ , then  $E_Q[f(Y^*, T^*, Z, X)|Y^*, T^*, X] = E_{Q_{Z|X}}[f(Y^*, T^*, Z, X)]$  for any  $f \in L^1(Q)$ .*

*Proof.* Let  $\mathcal{G}$  be the  $\sigma$ -field generated by  $(\mathcal{G}_{Y^*} \times \mathcal{G}_{T^*} \times \mathcal{G}_Z \times \mathcal{G}_X)$  where  $\mathcal{G}_V$  denotes the  $\sigma$ -field on which the marginal distribution  $Q_V$  is defined. We further define the class

$$\mathcal{A} \equiv \{A \in \mathcal{G} : E_Q[1\{(Y^*, T^*, Z, X) \in A\}|Y^*, T^*, X] = E_{Q_{Z|X}}[1\{(Y^*, T^*, Z, X) \in A\}]\}$$

and note that  $\mathbf{Y}^* \times \mathbf{T}^* \times \mathbf{Z} \times \mathbf{X} \in \mathcal{A}$ . Also observe that if  $A_1, A_2 \in \mathcal{A}$  and  $A_1 \subseteq A_2$  then  $A_2 \setminus A_1 \in \mathcal{A}$ . Next, let  $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$  be a sequence of pairwise disjoint sets and note

$$\begin{aligned} E_Q[1\{(Y^*, T^*, Z, X) \in \bigcup_{i=1}^\infty A_i\}|Y^*, T^*, X] &= \lim_{n \rightarrow \infty} E_{Q_{Z|X}}[\sum_{i=1}^n 1\{(Y^*, T^*, Z, X) \in A_i\}] \\ &= E_{Q_{Z|X}}[1\{(Y^*, T^*, Z, X) \in \bigcup_{i=1}^\infty A_i\}], \end{aligned} \quad (74)$$

by Theorem 10.1.5(4) in Bogachev (2007),  $\{A_i\}_{i=1}^\infty$  being disjoint, and  $A_i \in \mathcal{A}$  for all  $i$ . In particular, (74) implies  $\bigcup_{i=1}^\infty A_i \in \mathcal{A}$ , and we can therefore conclude that  $\mathcal{A}$  is a  $\lambda$ -system.

Let  $A_{Y^*} \in \mathcal{G}_{Y^*}$ ,  $A_{T^*} \in \mathcal{G}_{T^*}$ ,  $A_Z \in \mathcal{G}_Z$ , and  $A_X \in \mathcal{G}_X$  be arbitrary, and then observe

$$\begin{aligned} E_Q[1\{(Y^*, T^*, Z, X) \in A_{Y^*} \times A_{T^*} \times A_Z \times A_X\}|Y^*, T^*, X] \\ = E_{Q_{Z|X}}[1\{(Y^*, T^*, Z, X) \in A_{Y^*} \times A_{T^*} \times A_Z \times A_X\}] \end{aligned} \quad (75)$$

due to  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $Q$ . Result (75) implies  $A_{Y^*} \times A_{T^*} \times A_Z \times A_X \in \mathcal{A}$  and therefore that  $(\mathcal{G}_{Y^*} \times \mathcal{G}_{T^*} \times \mathcal{G}_Z \times \mathcal{G}_X) \subseteq \mathcal{A}$ . Since  $(\mathcal{G}_Z \times \mathcal{G}_X \times \mathcal{G}_{T^*} \times \mathcal{G}_{Y^*})$  is a  $\pi$ -system, the  $\pi - \lambda$  theorem (see, e.g., Theorem 2.38 in Pollard (2002)) implies  $\mathcal{A} = \mathcal{G}$ .

To conclude, let  $f \in L^1(Q)$  and  $\{f_n\}_{n=1}^\infty$  be a sequence of simple functions satisfying  $|f_n| \leq |f|$  and  $f_n(Y^*, T^*, Z, X) \rightarrow f(Y^*, T^*, Z, X)$  on a set with  $Q$ -probability one. By Proposition 10.1.7 in Bogachev (2007) we can then conclude that

$$\begin{aligned} E_Q[f(Y^*, T^*, Z, X)|Y^*, T^*, X] &= \lim_{n \rightarrow \infty} E_Q[f_n(Y^*, T^*, Z, X)|Y^*, T^*, X] \\ &= \lim_{n \rightarrow \infty} E_{Q_{Z|X}}[f_n(Y^*, T^*, Z, X)] = E_{Q_{Z|X}}[f(Y^*, T^*, Z, X)], \end{aligned}$$

where the second equality holds due to  $f_n$  being a simple function and  $\mathcal{A} = \mathcal{G}$ . ■

**LEMMA A.2:** *Let Assumptions 2.1 and 2.2 hold, and suppose that  $\mu(\pi(Z, X) > \delta) = 1$  for some  $\delta > 0$ . If  $\nu \in L^1(P)$ , then it follows  $\kappa = \nu/\pi$  satisfies  $\kappa \in L^1(P)$  and*

$$E_{\mu_{Z|X}}[\nu(Y^*(t), t, Z, X)1\{T^*(Z) = t\}] = E_{P_{Z|X}}[\kappa(Y^*(t), t, Z, X)1\{T^*(Z) = t\}].$$

*Proof.* Since  $\bar{Q} \ll \mu$  and  $\bar{Q}_{ZX} = P_{ZX}$  because  $\bar{Q} \in \Theta_0$ , it follows that  $\mu(\pi(Z, X) > \delta) = 1$  implies  $P(\pi(Z, X) > \delta) = 1$ . Hence,  $1/\pi \in L^\infty(P)$ , which together with  $\nu \in L^1(P)$  yields  $\kappa = \nu/\pi \in L^1(P)$ . The lemma therefore follows by definition of  $\kappa$  and  $\pi$ . ■

**LEMMA A.3:** *Let Assumptions 2.1, 2.2, and 4.1(ii) hold, and suppose that a finite collection  $\{s_j, Q_j\}_{j=1}^J$  with  $s_j \in \mathcal{S}_{Q_j}$  and  $Q_j \in \Theta_0$  for all  $1 \leq j \leq J$  satisfies*

$$P\left(\sum_{j=1}^J E_{Q_j}[s_j(Y^*, T^*, X)|Y, T, Z, X] = 0\right) = 1. \quad (76)$$

*Then, there is a  $Q^a \in \Theta_0$  and a  $\eta > 0$  such that the measure  $\tilde{Q}$  given by  $\tilde{Q}(A) \equiv Q^a(A) + \sum_{j=1}^J \eta E_{Q_j}[s_j(Y^*, T^*, X)1\{(Y^*, T^*, Z, X) \in A\}]$  belongs to  $\Theta_0$ .*

*Proof.* By Assumption 4.1(ii), there is a  $Q^i \in \Theta_0$  such that  $dQ^i/d\mu$  belongs to the interior of  $\mathcal{Q}$  in  $\mathbf{Q}$ . Set  $Q^a \equiv \lambda Q^i + (1 - \lambda) \sum_{j=1}^J Q_j/J$  and note  $dQ^a/d\mu$  belongs to the interior of  $\mathcal{Q}$  in  $\mathbf{Q}$  provided  $\lambda \in (0, 1)$  is chosen sufficiently large, and  $Q^a \in \Theta_0$  due to  $\Theta_0$  being convex by Lemma 4.1. Since  $\|s_j\|_{Q_j, \infty} < \infty$  for all  $1 \leq j \leq J$  because  $s_j \in \mathcal{S}_{Q_j} \subseteq L^\infty(Q_j)$ , the

definition of  $Q^a$  implies  $\tilde{Q}(A) \geq \sum_{j=1}^J Q_j(A)((1-\lambda)/J - \eta\|s_j\|_{Q_j, \infty})$ . Hence, since  $\lambda < 1$ ,  $\tilde{Q}$  is a positive measure provided we set  $\eta > 0$  sufficiently small. Further observe

$$\tilde{Q}(\mathbf{Y}^* \times \mathbf{T}^* \times \mathbf{Z} \times \mathbf{X}) = Q^a(\mathbf{Y}^* \times \mathbf{T}^* \times \mathbf{Z} \times \mathbf{X}) + \eta \sum_{j=1}^J E_{Q_j}[s_j(Y^*, T^*, X)] = 1 \quad (77)$$

by (76), the law of iterated expectations, and  $Q_j \in \Theta_0$ . It follows that  $\tilde{Q}$  is a probability measure. Moreover, since  $Q_j \ll \mu$  due to  $Q_j \in \Theta_0$ , we have  $\tilde{Q} \ll \mu$  and  $d\tilde{Q}/d\mu = dQ^a/d\mu + \eta \sum_j s_j dQ_j/d\mu$ . Therefore,  $dQ^a/d\mu$  belonging to the interior of  $\mathcal{Q}$  in  $\mathbf{Q}$  and  $s_j dQ_j/d\mu \in \mathbf{Q}$  by definition of  $\mathcal{S}_{Q_j}$  imply that  $d\tilde{Q}/d\mu \in \mathcal{Q}$  for  $\eta > 0$  small.

Next note that for any  $t \in \mathbf{T}$  and (measurable) set  $V$ , condition (76) and  $Q_j \in \Theta_0$  imply

$$\begin{aligned} & \sum_{j=1}^J E_{Q_j}[s_j(Y^*, T^*, X) 1\{T^*(Z) = t, (Y^*(t), Z, X) \in V\}] \\ &= \sum_{j=1}^J E_{Q_j}[E_{Q_j}[s_j(Y^*, T^*, X)|Y, T, Z, X] 1\{T = t, (Y, Z, X) \in V\}] = 0. \end{aligned} \quad (78)$$

Result (78),  $Q^a \in \Theta_0$ , and the definition of  $\tilde{Q}$  imply  $\tilde{Q}$  induces  $P$ . To conclude the proof, it remains to show  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $\tilde{Q}$ . Let  $f \in L^1(P_{ZX})$  and  $g$  be a bounded function of  $(Y^*, T^*, X)$ . Since  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $Q^a$  and  $Q_j$ , due to  $Q^a, Q_j \in \Theta_0$ , we have

$$\begin{aligned} E_{\tilde{Q}}[g(Y^*, T^*, X)f(Z, X)] &= E_{Q^a}[g(Y^*, T^*, X)E_{Q^a}[f(Z, X)|X]] \\ &\quad + \eta \sum_{j=1}^J E_{Q_j}[g(Y^*, T^*, X)s_j(Y^*, T^*, X)E_{Q_j}[f(Z, X)|X]]. \end{aligned} \quad (79)$$

However, since we showed  $\tilde{Q}$  is observationally equivalent to  $Q_0$  and  $Q^a$  and  $Q_j$  are observationally equivalent to  $Q_0$  (due to  $Q^a, Q_j \in \Theta_0$ ) it also follows that  $E_{Q^a}[f(Z, X)|X] = E_{Q_j}[f(Z, X)|X] = E_{\tilde{Q}}[f(Z, X)|X]$ . Combining this observation with (79) then yields

$$E_{\tilde{Q}}[g(Y^*, T^*, X)f(Z, X)] = E_{\tilde{Q}}[g(Y^*, T^*, X)E_{\tilde{Q}}[f(Z, X)|X]]. \quad (80)$$

Since (80) holds for any bounded  $g$  it follows  $E_{\tilde{Q}}[f(Z, X)|Y^*, T^*, X] = E_{\tilde{Q}}[f(Z, X)|X]$ ; see, e.g., Definition 10.1.1 in [Bogachev \(2007\)](#). Thus, since  $f \in L^1(P_{ZX})$  was also arbitrary, we conclude  $(Y^*, T^*) \perp\!\!\!\perp Z|X$  under  $\tilde{Q}$  and therefore that  $\tilde{Q} \in \Theta_0$ . ■

LEMMA A.4: Let Assumptions 2.1, 2.2 hold,  $\ell \in \bigcap_{Q \in \Theta_0} L^1(Q)$ ,  $\mathcal{S}_{\bar{Q}} = L^\infty(\bar{Q}_{Y^*T^*X})$ , and  $\mathcal{C} \subseteq \mathcal{R}$  be convex. If  $\ell$  belongs to the  $\tau$ -closure of  $\mathcal{C}$ , then there is a sequence  $\{f_n\}_{n=1}^\infty \subseteq L^1(\bar{Q}_{Y^*T^*X})$  such that  $\lim_{n \rightarrow \infty} \|\ell - f_n\|_{\bar{Q},1} = 0$ .

*Proof.* By Theorem 2.14 in Aliprantis and Border (2006) and  $\ell$  belonging to the  $\tau$ -closure of  $\mathcal{C}$ , there is a net  $\{c_\alpha\}_{\alpha \in \mathcal{A}} \subseteq \mathcal{C}$  such that  $\lim_\alpha \langle s, c_\alpha \rangle_{\bar{Q}} = \langle s, \ell \rangle_{\bar{Q}}$  for all  $s \in \mathcal{S}_{\bar{Q}} = L^\infty(\bar{Q}_{Y^*T^*X})$ . A second application of Theorem 2.14 in Aliprantis and Border (2006) implies  $\ell \in L^1(\bar{Q}_{Y^*T^*X})$  belongs to the closure of  $\mathcal{C} \subseteq L^1(\bar{Q}_{Y^*T^*X})$  under the weak topology generated by  $\mathcal{S}_{\bar{Q}} = L^\infty(\bar{Q}_{Y^*T^*X})$ . However, since  $L^\infty(\bar{Q}_{Y^*T^*X})$  is the norm dual of  $L^1(\bar{Q}_{Y^*T^*X})$  and  $\mathcal{C} \subseteq L^1(\bar{Q}_{Y^*T^*X})$  is convex, it follows that  $\ell$  belongs to the closure of  $\mathcal{C}$  under  $\|\cdot\|_{\bar{Q},1}$  (see, e.g., Theorem 3.12 in Rudin (1991)). Therefore, by Theorem 2.40(1) in Aliprantis and Border (2006),  $\|\ell - f_n\|_{\bar{Q},1} = o(1)$  for some  $\{f_n\}_{n=1}^\infty \subseteq \mathcal{C}$ . ■

LEMMA A.5: Let Assumptions 2.1 and 2.2 hold,  $\mathbf{T}^*$  be finite,  $\{C_i\}_{i=1}^r$  be a finite collection of subsets of  $\mathbf{T}^*$ , and define  $A : \bigotimes_{i=1}^r L^1(P_X) \rightarrow L^1(\bar{Q}_{T^*X})$  by setting  $A(f)(T^*, X) = \sum_{i=1}^r 1\{T^* \in C_i\} f_i(X)$  for any  $f = \{f_i\}_{i=1}^r \in \bigotimes_{i=1}^r L^1(P_X)$ . If  $\bar{Q}(T^* = t^*|X) \geq \varepsilon > 0$  a.s. for any  $t^* \in \mathbf{T}^*$  satisfying  $\bar{Q}(T^* = t^*) > 0$ , then the range of  $A$  is closed.

*Proof.* Let  $\mathbf{T}_0^* \equiv \{t^* \in \mathbf{T}^* : \bar{Q}(T^* = t^*) > 0\}$  and enumerate  $\mathbf{T}_0^*$  by  $\mathbf{T}_0^* = \{t_1^*, \dots, t_{d^*}^*\}$ . Interpret any  $\{f_i\}_{i=1}^r \in \bigotimes_{i=1}^r L^1(P_X)$  as a column vector  $f(X) \equiv (f_1(X), \dots, f_r(X))'$  and define a  $d^* \times r$  matrix  $\Omega$  whose  $j^{\text{th}}$  row is given by  $\omega_j \equiv (1\{t_j^* \in C_1\}, \dots, 1\{t_j^* \in C_r\})$ . Then note that  $\bar{Q}(T^* = t^*|X) \geq \varepsilon > 0$  for all  $t^* \in \mathbf{T}_0^*$  and  $\bar{Q}_X = P_X$  due to  $\bar{Q} \in \Theta_0$  imply

$$E_{\bar{Q}}[|A(f)(T^*, X)|] \geq \varepsilon E_{P_X}[\|\Omega f(X)\|_1] \geq \varepsilon \eta E_{P_X}[\|\Omega^\dagger \Omega f(X)\|_1], \quad (81)$$

where the final inequality holds for some  $\eta > 0$  and  $\Omega^\dagger$  the Moore-Penrose pseudoinverse of  $\Omega$  because  $\Omega \Omega^\dagger \Omega = \Omega$  and  $\Omega : \text{range}\{\Omega^\dagger\} \rightarrow \mathbf{R}^{d^*}$  is invertible. Next suppose  $\|A(f_n) - \ell\|_{\bar{Q},1} = o(1)$  for some  $\{f_n\}_{n=1}^\infty \in \bigotimes_{i=1}^r L^1(P_X)$  and  $\ell \in L^1(\bar{Q}_{T^*X})$ . Result (81) implies  $\{\Omega^\dagger \Omega f_n\}_{n=1}^\infty$  is Cauchy in  $\bigotimes_{i=1}^r L^1(P_X)$  and therefore, by completeness of  $\bigotimes_{i=1}^r L^1(P_X)$ , there is  $\tilde{f} \in \bigotimes_{i=1}^r L^1(P_X)$  satisfying  $E_{P_X}[\|\Omega^\dagger \Omega f_n(X) - \tilde{f}(X)\|_1] = o(1)$ . The same manipulations as in result (81) and  $\Omega = \Omega \Omega^\dagger \Omega$  then imply that

$$E_{\bar{Q}}[\|\ell(T^*, X) - A(\tilde{f})(T^*, X)\|] \leq \lim_{n \rightarrow \infty} E_{\bar{Q}}[\|A(f_n - \tilde{f})(T^*, X)\|]$$

$$\leq \lim_{n \rightarrow \infty} E_{P_X} [\|\Omega f_n(X) - \Omega \tilde{f}(X)\|_1] \leq \lim_{n \rightarrow \infty} \|\Omega\|_{o,1} E_{P_X} [\|\Omega^\dagger \Omega f_n(X) - \tilde{f}(X)\|_1] = 0,$$

where  $\|\cdot\|_{o,1}$  denotes the operator norm of  $\Omega : (\mathbf{R}^r, \|\cdot\|_1) \rightarrow (\mathbf{R}^{d^*}, \|\cdot\|_1)$  and the equality follows from  $E_{P_X} [\|\Omega^\dagger \Omega f_n(X) - \tilde{f}(X)\|_1] = o(1)$ . Hence, the range of  $A$  is closed. ■

## APPENDIX B: PROOFS FOR SECTION 5

*Proof of Theorem 5.1.* The first equality in (38) follows from the definitions of  $\hat{\lambda}$  and  $\hat{\psi}_k$  and applying Lemma B.1 with  $W_i$  satisfying  $P(W_i = 1) = 1$ . Since  $\|\psi\|_\infty \lesssim B$  by Assumption 5.2(i) and  $\sigma^2 = \text{Var}_P\{\psi(Y, T, Z, X)\}$  by definition, Theorem 1.1 in Zhai (2018) yields

$$\frac{1}{\sqrt{n}\sigma} \sum_{i=1}^n (\psi(Y_i, T_i, Z_i, X_i) - E_P[\psi(Y, T, Z, X)]) = \mathbb{Z} + O_P\left(\frac{B \log(n)}{\sigma \sqrt{n}}\right) \quad (82)$$

for  $\mathbb{Z}$  a standard normal random variable possibly depending on  $n$ . The second equality in (38) then follows from (82), Lemma B.2, and  $B \log(n) = o(\sigma \sqrt{n})$  by Assumption 5.2(i). ■

*Proof of Theorem 5.2.* First note that Lemma B.1,  $(\hat{\lambda} - \lambda_{Q_0}) = O_P(\sigma/\sqrt{n})$  by Theorem 5.1, and  $\sum_i W_i/n = o_P(1)$  due to  $E[W] = 0$  together allow us to conclude that

$$\frac{\sqrt{n}}{\sigma} (\hat{\lambda}^* - \hat{\lambda}) = \frac{1}{\sqrt{n}\sigma} \sum_{i=1}^n W_i \psi(Y_i, T_i, Z_i, X_i) + o_P(1). \quad (83)$$

Next note that  $\sigma^2 \equiv \text{Var}_P\{\psi(Y, T, Z, X)\}$  by definition,  $\|\psi\|_\infty \lesssim B$  and  $B \log(n) = o(\sigma \sqrt{n})$  by Assumption 5.2(i), and Bernstein's inequality yield that for  $n$  sufficiently large

$$P\left(\left|\frac{1}{n\sigma} \sum_{i=1}^n (\psi(Y_i, T_i, Z_i, X_i) - E_P[\psi(Y, T, Z, X)])\right| > \frac{\log(n)}{\sqrt{n}}\right) \leq 2 \exp\left\{-\frac{\log^2(n)}{4}\right\}. \quad (84)$$

In particular, note that result (84) together with Lemma B.2 allow us to conclude that

$$\left|\frac{1}{n} \sum_{i=1}^n \{\psi^2(Y_i, T_i, Z_i, X_i) - (\psi(Y_i, T_i, Z_i, X_i) - E_P[\psi(Y, T, Z, X)])^2\}\right| = o_P\left(\frac{\sigma^2 \log(n)}{n}\right). \quad (85)$$

Moreover, Markov's inequality,  $\|\psi\|_\infty \lesssim B$ , and  $\sigma^2 \equiv \text{Var}_P\{\psi(Y, T, Z, X)\}$  further imply

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n \frac{(\psi(Y_i, T_i, Z_i, X_i) - E_P[\psi(Y, T, Z, X)])^2}{\sigma^2} - 1\right| > \epsilon\right) \lesssim \frac{B^2 \sigma^2}{n \sigma^4 \epsilon^2} \quad (86)$$

for any  $\epsilon > 0$ . Therefore, setting  $\hat{\sigma}^2 \equiv \sum_i \psi^2(Y_i, T_i, Z_i, X_i)/n$ , we can conclude from results (85) and (86) together with  $B \log(n) = o(\sigma\sqrt{n})$  by Assumption 5.2(i) that

$$\frac{\hat{\sigma}^2}{\sigma^2} = \frac{1}{n} \sum_{i=1}^n \frac{(\psi(Y_i, T_i, Z_i, X_i) - E_P[\psi(Y, T, Z, X)])^2}{\sigma^2} + o_P(1) = 1 + o_P(1). \quad (87)$$

To conclude, note that  $\{W_i\}_{i=1}^n$  being i.i.d. standard normal random variables and  $\{W_i\}_{i=1}^n$  being independent of  $\{Y_i, T_i, Z_i, X_i\}_{i=1}^n$  imply  $\mathbb{Z}^* \equiv (n\hat{\sigma}^2)^{-1/2} \sum_{i=1}^n W_i \psi(Y_i, T_i, Z_i, X_i)$  satisfies  $\mathbb{Z}^* \sim N(0, 1)$  conditionally on  $\{Y_i, T_i, Z_i, X_i\}_{i=1}^n$  and hence  $\mathbb{Z}^*$  is independent of  $\{Y_i, T_i, Z_i, X_i\}_{i=1}^n$ . The theorem therefore follows from (83) and (87). ■

**LEMMA B.1:** *Let Assumptions 5.1(i)(iii) and 5.2(i)(ii)(v) hold,  $\{W_i\}_{i=1}^n$  be an i.i.d. sequence independent of  $\{Y_i, T_i, Z_i, X_i\}_{i=1}^n$  satisfying  $E[W^2] < \infty$ ,  $\psi$  and  $\hat{\psi}_k$  be as defined in (37) and (39) respectively, and  $\sigma^2 \equiv \text{Var}_P\{\psi(Y, T, Z, X)\}$ . Then it follows that*

$$\frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} W_i \{\hat{\psi}_k(Y_i, T_i, Z_i, X_i) + \hat{\lambda}\} = \frac{1}{n} \sum_{i=1}^n W_i \{\psi(Y_i, T_i, Z_i, X_i) + \lambda_{Q_0}\} + o_P\left(\frac{\sigma}{\sqrt{n}}\right).$$

*Proof.* Let  $\hat{\Delta}_k^\gamma \equiv (\hat{\gamma}_k - \gamma)$ ,  $\hat{\Delta}_k^\beta \equiv (\hat{\beta}_k - \beta)$ , and note for any  $k$  we have the decomposition

$$\begin{aligned} & \frac{1}{n} \sum_{i \in I_k} W_i \{(\hat{\psi}_k(Y_i, T_i, Z_i, X_i) + \hat{\lambda}) - (\psi(Y_i, T_i, Z_i, X_i) + \lambda_{Q_0})\} \\ &= \frac{1}{n} \sum_{i \in I_k} W_i \{(\nu_j(Y_i, T_i, Z_i, X_i) - b(Z_i, X_i)' \beta) b(Z_i, X_i)' \hat{\Delta}_k^\gamma \\ &+ \frac{1}{n} \sum_{i \in I_k} W_i \{E_{\mu_{Z|X}}[b(Z, X_i)'] - b(Z_i, X_i)' \gamma b(Z_i, X_i)'\} \hat{\Delta}_k^\beta \\ &- \frac{1}{n} \sum_{i \in I_k} W_i (\hat{\Delta}_k^\gamma)' b(Z_i, X_i) b(Z_i, X_i)' \hat{\Delta}_k^\beta. \end{aligned} \quad (88)$$

Next, note that  $E_P[(\nu_j(Y, T, Z, X) - b(Z, X)' \beta) b(Z, X)] = 0$  as a result of the first order conditions for  $\beta$ . Hence, since  $\hat{\gamma}_k$  is computed using the observations in  $I_k^c$ , we obtain

$$E\left[\frac{1}{n} \sum_{i \in I_k} W_i (\nu_j(Y_i, T_i, Z_i, X_i) - b(Z_i, X_i)' \beta) b(Z_i, X_i)' \hat{\Delta}_k^\gamma | \hat{\gamma}_k\right] = 0.$$

Therefore, the observations within  $I_k$  being i.i.d.,  $|I_k| \asymp n$  by Assumption 5.2(v),  $\|b'\beta\|_\infty \vee \|\nu_j\|_\infty \equiv B < \infty$  by Assumption 5.2(i),  $E[W^2] < \infty$ , and the definition of  $r^\gamma$  yield

$$\text{Var}\left\{\frac{1}{n} \sum_{i \in I_k} W_i(\nu_j(Y_i, T_i, Z_i, X_i) - b(Z_i, X_i)' \beta)) b(Z_i, X_i)' \hat{\Delta}_k^\gamma | \{\hat{\gamma}_{t,k}\}_{t \in \mathbf{T}}\right\} \lesssim \frac{B^2 r_\gamma^2}{n}. \quad (89)$$

Moreover, by identical arguments but relying on the first order condition for  $\gamma$  yields that

$$\begin{aligned} & \text{Var}\left\{\frac{1}{n} \sum_{i \in I_k} W_i \{E_{\mu_{Z|X}}[b(Z, X_i)'] - b(Z_i, X_i)' \gamma b(Z_i, X_i)'\} \hat{\Delta}_k^\beta | \hat{\beta}_k\right\} \\ & \lesssim \frac{1}{n} \{E_P[(E_{\mu_{Z|X}} b(Z, X)' \hat{\Delta}_k^\beta)^2] + E_P[(b(Z, X)' \gamma)^2 (b(Z, X)' \hat{\Delta}_k^\beta)^2]\} \lesssim \frac{r_\beta^2}{n}, \end{aligned} \quad (90)$$

where in the final inequality we employed Jensen's inequality, Assumptions 5.1(iii) and 5.2(i), and  $d\mu_{Z|X}/dP_{Z|X} = 1/\pi$ . Finally, note that the Cauchy-Schwarz inequality yields

$$\begin{aligned} & \left| \frac{1}{n} \sum_{i \in I_k} W_i (\hat{\Delta}_k^\gamma)' b(Z_i, X_i) b(Z_i, X_i)' \hat{\Delta}_k^\beta \right| \\ & \leq \left\{ \frac{1}{n} \sum_{i \in I_k} W_i^2 (b(Z_i, X_i)' \hat{\Delta}_k^\gamma)^2 \right\}^{1/2} \times \left\{ \frac{1}{n} \sum_{i \in I_k} (b(Z_i, X_i)' \hat{\Delta}_k^\beta)^2 \right\}^{1/2}. \end{aligned} \quad (91)$$

The lemma follows from results (88), (89), (90), and (91),  $E[W_i^2] < \infty$ ,  $\{W_i\}_{i=1}^n$  being independent of  $\{Y_i, T_i, Z_i, X_i\}_{i=1}^n$ , Markov's inequality and Assumption 5.2(ii). ■

**LEMMA B.2:** *If Assumptions 5.1(ii), 5.2(iii)(iv) hold, and  $\psi$  is as defined in (37), then it follows that  $|E_P[\psi(Y, T, Z, X)]| = o(\sigma/\sqrt{n})$ .*

*Proof.* First, note that the first order conditions for  $\gamma$  imply  $E_{P_X}[E_{\mu_{Z|X}}[b(Z, X)' \beta]] - E_P[(b(Z, X)' \gamma)(b(Z, X)' \beta)] = 0$ . Together with the definition of  $\psi$ , the law of iterated expectations, and  $\kappa_j = \nu_j/\pi$ , this first order condition therefore implies that

$$\begin{aligned} & |E_P[\psi(Y, T, Z, X)] - (E_P[\kappa_j(Y, T, Z, X)] - \lambda_{Q_0})| \\ & = |(E_P[(b(Z, X)' \gamma - \frac{1}{\pi(Z, X)})](E_P[\nu_j(Y, T, Z, X)|Z, X] - b(Z, X)' \beta))|. \end{aligned} \quad (92)$$

Therefore, the claim of the lemma follows from result (92), the Cauchy-Schwarz inequality, and Assumptions 5.2(iii)(iv). ■

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